Calibeating Prediction-Powered Inference

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Abstract

We study semisupervised mean estimation with a small labeled sample, a large unlabeled sample, and an available black-box prediction model that may be miscalibrated. A standard approach in this setting is augmented inverse-probability weighting (AIPW) [81], which is robust to prediction-model misspecification but may be inefficient when the prediction score is poorly calibrated. We propose *Calibrated Prediction-Powered Inference*: post-hoc calibrate the prediction score on the labeled sample and then average the calibrated predictions over the pooled sample. This simple post-hoc step requires no retraining and can improve efficiency relative to using the original score directly. We study both linear and isotonic calibration. For isotonic calibration, we prove a first-order *calibeating* guarantee: isotonic post-processing can improve on the original score and on simpler post-processing rules in both predictive accuracy and estimator efficiency, and no further post-processing of the fitted isotonic score yields additional first-order improvement. For linear calibration, we show first-order equivalence to PPI++. We also show that the original PPI estimator is a special case of AIPW and can be inefficient when the prediction model is accurate, while PPI++ is AIPW with empirical-efficiency maximization [90]. In simulations and real-data experiments, our calibrated estimators often outperform PPI and remain competitive with, or outperform, AIPW and PPI++. We provide an accompanying `Python` package, `ppi_aipw`.

1 Introduction

We study semisupervised mean estimation with one labeled dataset $\{(X_i, Y_i)\}_{i=1}^n$ and one unlabeled dataset $\{\widetilde{X}_i\}_{i=1}^N$, where outcomes are observed only for a subset sampled completely at random. This is a classical missing-outcome problem: rich covariate information is available for many individuals, but the outcome of interest is observed only for a smaller subset. Such settings arise when gold-standard outcomes are expensive but prediction scores are readily available, for example in randomized trials with prognostic models trained on historical data [96], in biobank analyses using phenotype predictors built from EHR or omics data [47, 128], and in other data-rich pipelines that supplement costly measurements with model-based predictions [1, 130]. This problem has long been studied in missing-data and semiparametric statistics, where methods such as AIPW, TMLE, and related debiased estimators allow flexible machine-learning predictors to be used with valid inference [81, 114, 116, 115, 22]. More recently, related ideas reappeared in the machine-learning literature under the name prediction-powered inference [1, 2, 130] and in covariate adjustment for randomized trials [40, 90, 66, 64, 96, 26, 50].

A standard approach is to use the labeled sample to debias a plug-in estimator based on a prediction model. If $m(X)$ is a black-box prediction score, the plug-in estimator averages $m(X)$ over the pooled covariate sample, while the labeled sample estimates the residual correction term $\mathbb{E}\{Y - m(X)\}$. The AIPW estimator [81] averages $m(X)$ over both labeled and unlabeled samples and

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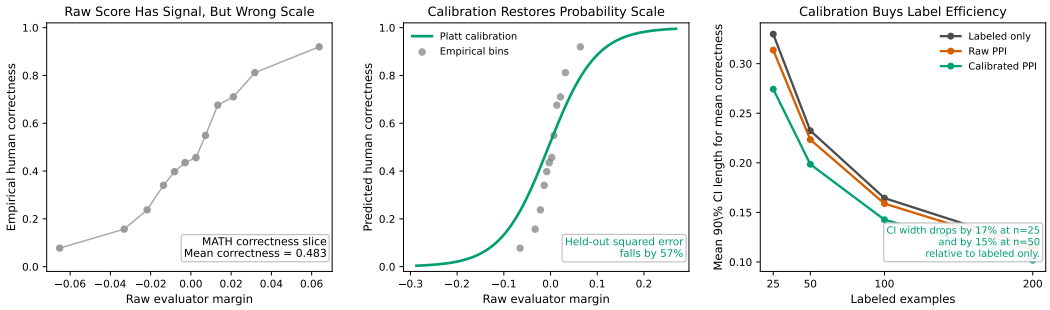

Figure 1: Grounded one-mean toy derived from the MATH slice of a public LLM-evaluation benchmark with human correctness labels, using ArmoRM as the automatic evaluator score. Left: the raw evaluator margin has clear signal about correctness, but it is not on the right numerical scale. Middle: a Platt map fit on 100 labeled examples restores the probability scale and lowers held-out squared error by about 58%. Right: in Monte Carlo with $N = 2000$ unlabeled scores, calibrated PPI yields about 17% shorter 90% Wald intervals than labeled-only estimation at $n = 25$ and about 15% shorter intervals at $n = 50$, while raw PPI provides only modest gains.

then adds the average residual over the labeled sample, whereas the PPI estimator [1] averages $m(X)$ only over the unlabeled covariates before applying the same residual correction. PPI can therefore be less efficient because it discards labeled-covariate information. These methods protect against bias from imperfect prediction models, but they usually treat the prediction score itself as fixed. In practice, however, useful black-box prediction scores are often systematically too high, too low, or too extreme because of population mismatch, temporal drift, distribution shift, or regularization bias.

This motivates post-hoc calibration of the prediction score. Rather than correcting only the marginal bias $\mathbb{E}\{Y - m(X)\}$, one can use the labeled sample to transform the score so that its numerical values better align with the outcomes they are meant to predict [126, 127, 70]. An ideal target is a calibrated score $m_n^\star(X)$ satisfying $\mathbb{E}[Y \mid m_n^\star(X)] = m_n^\star(X)$, in which case $\mathbb{E}[m_n^\star(X)] = \mathbb{E}[Y]$ and the corresponding plug-in estimator is automatically unbiased for the population mean. This is attractive because a useful prediction model may already be available, while a relatively small labeled sample can calibrate its output without retraining [86]. Classical post-processing methods include Platt scaling, linear calibration, histogram binning, and isotonic regression [75, 65, 126, 127]. Such calibration is especially relevant for modern machine-learning predictors, which are often informative yet systematically miscalibrated [7, 34, 122].

An early grounded toy example makes the mechanism concrete. We take a public LLM-evaluation benchmark with human correctness labels, focus on its MATH slice, and ask how accurately one can estimate average human correctness from a small labeled sample plus many unlabeled evaluator scores. The raw evaluator margin has real signal but is poorly scaled for the human labels, so using it directly in PPI yields only modest gains. A one-dimensional recalibration fit on the labeled pairs restores the probability scale, sharply reduces held-out squared error, and yields materially shorter confidence intervals for the mean. In this sense, calibration improves label efficiency: achieving the same uncertainty with the raw score or with labeled-only estimation would require more human labels. Figure 1 visualizes this mechanism in the resulting one-mean toy.

Our proposal is simple: fit a black-box regression model, calibrate it on the labeled sample, and average the calibrated predictions over the pooled covariate sample. We call this workflow *Calibrated Prediction-Powered Inference*. It remains a simple "predict, calibrate, and average" procedure, but it also admits an exact debiased interpretation: the resulting estimator is exactly equal to the AIPW estimator based on the calibrated score, with the residual correction encoded by the calibration step itself. Because calibration can improve the score, it can also improve efficiency relative to AIPW based on the uncalibrated model.

Contributions. We make three main contributions. First, we study *calibrated prediction-powered inference* for semisupervised mean estimation, covering familiar post-hoc methods such as linear and isotonic calibration. Second, we show that calibration can improve the prediction score and therefore

precision while also making the pooled plug-in estimator automatically debiased because it is exactly equal to AIPW with the calibrated score. Third, we develop asymptotic theory: isotonic-calibrated PPI gives a lightweight default with valid inference, efficiency gains under miscalibration, and no further first-order gain from post-processing, while linearly calibrated PPI is first-order equivalent to PPI++ and related to classical prognostic-score regression adjustment.

1.1 Related work

Although PPI has been presented as a recent development, the underlying problem and many of its estimators sit inside a broader literature on semiparametric debiasing and regression adjustment.

Semiparametric statistics and missing data. Rubin formalized missing at random as a key identification condition for recovering full-data targets from partially observed outcomes [91], and semiparametric efficiency theory then characterized efficient influence functions, regular asymptotically linear estimators, and one-step debiased procedures [81, 82, 80, 114, 101, 10]. This yielded the influence-function and efficiency framework used throughout the paper.

Debiased machine learning and targeted learning. A large body of work combines this semiparametric framework with machine-learning-based nuisance estimation under the labels of targeted learning and debiased machine learning [116, 115, 22, 24]. In the present setting, these methods already imply that flexible black-box predictors can be used with valid inference once the missingness probability is known.

Semisupervised inference and PPI. Related work studies semisupervised or surrogate-assisted estimation when the primary outcome is observed only on a limited subset [72, 73, 19, 18, 21, 49, 52]. More recent machine-learning papers popularized the label *prediction-powered inference* (PPI) for this semisupervised mean-estimation setting [1, 2, 130]. Methodologically, however, PPI estimators are best understood as semiparametric variants: PPI++ is AIPW with empirical-efficiency maximization [90, 114], MEC [57] uses generalized entropy weights to balance the prediction score, and closely related questions have long been studied in randomized trials [40, 84, 66, 67, 96, 46]. Here we emphasize the corresponding regression-adjustment view.

Calibrated DML and Recalibrated PPI. In our mean-estimation setting, the closest precursor is the calibrated debiased machine-learning framework of van der Laan et al. [108], which studies calibration of the outcome regression in doubly robust inference. Our problem can be viewed as a semisupervised special case of that calibrated-DML perspective, which also motivates the term *calibrated PPI*. A closely related recent paper is Ji et al. [49], who study recalibrated prediction-powered inference for general estimating equations. In the mean-estimation setting, their method effectively learns the oracle correction to the initial prediction score that gives the most efficient prediction-powered estimator. In the squared-loss case, this is equivalent to learning the oracle conditional-mean recalibration of the initial score, so the two approaches are closely related. The difference is in emphasis and theory. Ji et al. [49] focus on learning the oracle correction for efficiency. By contrast, we focus on calibrating the score itself and on what can be guaranteed from simple post-hoc calibration rules, even without consistently learning the oracle target. This yields lightweight plug-in methods that can be added to an existing prediction pipeline without retraining, together with calibration-specific results: an exact AIPW interpretation, first-order equivalence between linear calibration and PPI++, and a first-order calibeating guarantee for isotonic calibration.

From missing data to causal inference. From the perspective of estimation and efficiency theory, semisupervised mean estimation with missing-at-random outcomes is statistically equivalent to randomized experiments and related sampling problems once the missingness indicator is reinterpreted as a treatment or sampling indicator [92, 114, 30, 69]. This lets tools from covariate adjustment, causal inference, and model-assisted estimation transfer directly.

Calibration. Classical post-processing methods such as Platt scaling, linear calibration, isotonic regression, and histogram binning improve predictive reliability without retraining the underlying model [75, 25, 36, 65, 126, 127, 70, 35]. Recent applications also include value prediction in offline reinforcement learning [104, 110].

2 Two-Sample Semisupervised Setup

2.1 Data structure and notation

We observe two independent datasets,

$$\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n \qquad \text{and} \qquad \mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N.$$

The labeled sample consists of i.i.d. draws from a distribution P_0 on $\mathcal{X} \times \mathbb{R}$, while the unlabeled sample consists of i.i.d. draws from the distribution $P_{0,X}$ of X under P_0. Our target is the population mean $\psi_0 := \mathbb{E}_{P_0}[Y]$, and we write $\mu_0(x) := \mathbb{E}_{P_0}[Y \mid X = x]$.

For functions $a : \mathcal{X} \times \mathbb{R} \to \mathbb{R}$ and $b : \mathcal{X} \to \mathbb{R}$, define the empirical means over the labeled and unlabeled samples by

$$\mathbb{P}_n^L a := \frac{1}{n}\sum_{i=1}^n a(X_i, Y_i), \qquad \mathbb{P}_N^U b := \frac{1}{N}\sum_{j=1}^N b(\widetilde{X}_j).$$

For readability, we will sometimes write $\mathbb{P}_n^L\{a(X,Y)\}$ instead of $\mathbb{P}_n^L a$, and for functions of X alone we write $\mathbb{P}_n^L\{b(X)\} = n^{-1}\sum_{i=1}^n b(X_i)$. Let $M := n + N$ and $\rho_n := n/M$; pooled score averages are written as $\rho_n \mathbb{P}_n^L f + (1-\rho_n)\mathbb{P}_N^U f$.

Assumption 1 (Two-sample design)**.** The following hold:

(i) $\mathcal{D}_L$ is i.i.d. from P_0, $\mathcal{D}_U$ is i.i.d. from $P_{0,X}$, and the two samples are independent.

(ii) Y has finite second moment under P_0.

(iii) With $M := n + N$, the labeled fraction $\rho_n = n/M$ satisfies $\rho_n \to \rho_0$ for some $\rho_0 \in (0,1)$.

2.2 Review of AIPW, PPI, and semiparametric efficiency

The AIPW class. We begin with the augmented inverse-probability weighted (AIPW) estimator in the unrestricted two-sample model [81]. It is helpful to view AIPW not as a single estimator, but as a family of estimators indexed by a score function $f : \mathcal{X} \to \mathbb{R}$ with finite second moment. For any such f, define

$$\widehat{\psi}(f) := \rho_n \mathbb{P}_n^L\{f(X)\} + (1-\rho_n)\mathbb{P}_N^U\{f(\widetilde{X})\} + \mathbb{P}_n^L\{Y - f(X)\}. \tag{1}$$

This estimator combines a plug-in term that imputes Y from $f(X)$ with a residual correction from the labeled sample. For any fixed f, $\widehat{\psi}(f)$ is unbiased in finite samples for ψ_0. Thus, the choice of f affects efficiency through variance, but not validity. This makes AIPW a safe estimator [124] and a particularly simple instance of double robustness [5]; it is the semisupervised version of the broader covariate-adjustment logic used in randomized trials [66, 67, 96, 46].

Asymptotic linearity and efficiency. Appendix B gives the full influence-pair characterization of this class and the efficient influence pair in the unrestricted two-sample model. For the main paper, the key consequence is that the asymptotic variance of $\sqrt{M}\{\widehat{\psi}(f) - \psi_0\}$ can be written as

$$\mathrm{AVar}\Big\{\sqrt{M}\,(\widehat{\psi}(f) - \psi_0)\Big\} = V_{\text{eff}} + \frac{1-\rho_0}{\rho_0}\mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big],$$

where $\tilde{f}(X) := f(X) - \mathbb{E}_{P_{0,X}}\{f(X)\} + \psi_0$ is the recentered score and V_{eff} is the semiparametric efficiency bound [10]. Thus, within the AIPW class, efficiency is governed by the mean-squared error of the recentered score $\tilde{f}$ relative to $\mu_0(X)$, and the variance is minimized when $\tilde{f} = \mu_0$, in particular at $f = \mu_0$.

AIPW, PPI, and empirical-efficiency maximization. The usual AIPW estimator corresponds to $f(X) = m(X)$, where $m(X)$ is a prediction score. The original PPI estimator of Angelopoulos et al. [1] corresponds to the scaled score choice $f(X) = m(X)/(1-\rho_n)$, so it is another score choice in the same AIPW class. Relative to AIPW, PPI places the plug-in term entirely on the unlabeled sample and therefore discards labeled-sample covariate information, so it can be less efficient when the original score is already informative.

Because both AIPW and PPI lie in the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, it is natural to choose λ data-adaptively rather than fixing it in advance. This yields the estimator called PPI++ by [2]: the AIPW estimator with the empirically rescaled score $\hat{\lambda} m(X)$. Here $\hat{\lambda}$ is chosen by minimizing the empirical variance criterion over $f(X) = \lambda m(X)$, a procedure known as empirical-efficiency maximization [90, 114], and it targets

$$\lambda^\star = \frac{\mathrm{Cov}\big(Y, m(X)\big)}{\mathrm{Var}\big(m(X)\big)}. \tag{2}$$

The same principle extends beyond simple rescaling: one can minimize the same criterion over richer classes, such as $\{\lambda m(X) + \beta^\top X\}$, monotone score transformations, or an RKHS, and can use it in cross-validation to choose among candidate estimator classes [90]; see Appendix D. For the main paper, the point is simple: AIPW remains safe across the whole class, and our proposal is to improve the score by calibration rather than to change the underlying validity mechanism.

3 Calibrated Prediction-Powered Inference

Section 2 showed that AIPW is safe while the choice of score $f(X)$ determines efficiency. We now introduce calibrated prediction-powered inference, which improves an available prediction score $m(X)$ through post-processing rather than retraining. Using the labeled sample, we learn a transformation of $m(X)$, average the calibrated predictions over the pooled sample, and obtain an estimator that has a simple plug-in form but is exactly AIPW.

3.1 General estimator and exact AIPW representation

Let $m : \mathcal{X} \to \mathbb{R}$ denote a black-box regression model. A calibration procedure uses the labeled scores $m(X_1), \ldots, m(X_n)$ and outcomes $Y_1, \ldots, Y_n$ to fit a post-hoc map f_n, and then defines the calibrated regression model

$$m_n^\star := f_n \circ m.$$

The resulting predictions on the labeled and unlabeled samples are $m_n^\star(X_i)$ for $i = 1, \ldots, n$ and $m_n^\star(\widetilde{X}_j)$ for $j = 1, \ldots, N$. Intuitively, the map f_n updates the original score $m(X)$ so that its numerical values better align with the outcome.

This post-processing step has two main goals. First, it aims to improve the original prediction score m through a simple transformation, without retraining the underlying model, ideally by improving calibration without sacrificing predictive performance. Second, it is designed so that the resulting plug-in estimator is automatically unbiased, without requiring an explicit residual correction.

Given the post-processed score $m_n^\star$, define the calibrated plug-in estimator by

$$\widehat{\psi}_{\mathrm{cal}} := \frac{1}{n+N} \left\{ \sum_{i=1}^{n} m_n^\star(X_i) + \sum_{j=1}^{N} m_n^\star(\widetilde{X}_j) \right\}. \tag{3}$$

Equivalently, $\widehat{\psi}_{\mathrm{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\}$.

To distinguish how the post-processing map is fit from what property the resulting score satisfies, we introduce two function classes. Let $\mathcal{H}$ be an optimization class of real-valued maps $h : \mathbb{R} \to \mathbb{R}$, and let $\mathcal{F}$ be a linear calibration class of real-valued maps $f : \mathbb{R} \to \mathbb{R}$ that contains the constant functions and the identity map. We fit the post-processing map f_n by empirical risk minimization over $\mathcal{H}$, in the sense that

$$f_n \in \arg\min_{h \in \mathcal{H}} \mathbb{P}_n^L\big[(Y - h(m(X)))^2\big].$$

We then ask what property the resulting post-processed score $m_n^\star := f_n \circ m$ satisfies relative to the calibration class $\mathcal{F}$.

We say that $m_n^\star$ is *empirically calibrated over* $\mathcal{F}$ if, for every $f \in \mathcal{F}$,

$$\mathbb{P}_n^L\big[(Y - m_n^\star(X))^2\big] \le \mathbb{P}_n^L\big[(Y - f(m_n^\star(X)))^2\big], \tag{4}$$

where $\mathbb{P}_n^L$ denotes the empirical mean over the labeled sample. In words, once $m_n^\star$ has been calibrated, no further transformation of $m_n^\star$ within $\mathcal{F}$ can improve empirical squared-error fit on the labeled sample.

Equivalently, the first-order optimality conditions for (4) are

$$\mathbb{P}_n^L\big[f(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = 0, \qquad f \in \mathcal{F}. \tag{5}$$

Because $\mathcal{F}$ contains the constant functions and the identity map, these conditions imply, in particular,

$$\mathbb{P}_n^L\{Y - m_n^\star(X)\} = 0, \qquad \mathbb{P}_n^L\big[m_n^\star(X)\{Y - m_n^\star(X)\}\big] = 0.$$

The first identity makes the AIPW residual term vanish exactly on the labeled sample.

Thus a calibration method is naturally described by a pair $(\mathcal{H}, \mathcal{F})$: linear and fixed-bin histogram calibration have $\mathcal{H} = \mathcal{F}$, whereas isotonic regression uses a monotone $\mathcal{H}$ but, by Proposition 1, yields a much larger $\mathcal{F}$.

The next result gives the basic structural property of calibrated PPI estimators.

Theorem 1 (Exact AIPW representation under empirical calibration)**.** *Suppose the calibrated score satisfies equation (5), where $\mathcal{F}$ contains all affine maps. Then, for any $\widehat{w} \in \mathcal{F}$,*

$$\widehat{\psi}_{\text{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\} + \mathbb{P}_n^L\big[\widehat{w}(m_n^\star(X))\{Y - m_n^\star(X)\}\big]. \tag{6}$$

In particular, taking $\widehat{w} \equiv 1$ yields the exact AIPW representation

$$\widehat{\psi}_{\text{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m_n^\star(X)\}.$$

If, in addition, $\widehat{w}_n^\star \in \mathcal{F}$ satisfies

$$\mathbb{P}_n^L\big[\widehat{w}_n^\star(m_n^\star(X))\, f(m_n^\star(X))\big] = \rho_n \mathbb{P}_n^L\{f(m_n^\star(X))\} + (1-\rho_n)\mathbb{P}_N^U\{f(m_n^\star(\widetilde{X}))\}, \qquad f \in \mathcal{F}, \tag{7}$$

then

$$\widehat{\psi}_{\text{cal}} = \mathbb{P}_n^L\big[\widehat{w}_n^\star(m_n^\star(X))Y\big]. \tag{8}$$

Theorem 1 turns the pooled plug-in rule into exact AIPW with the calibrated score and, when (7) holds, into an equivalent balancing-weight estimator.

Full versus linear calibration. In *full calibration*, $\mathcal{F}$ is the class of all real-valued transformations, while $\mathcal{H}$ is the fitting class used to learn f_n. This distinction matters because $\mathcal{H}$ regularizes the fit: without it, one could overfit the labeled sample and achieve perfect empirical fit with poor population prediction. In practice, $\mathcal{H}$ may consist of piecewise-constant or monotone maps, as in histogram and isotonic calibration. In *linear calibration*, by contrast, $\mathcal{H} = \mathcal{F}$ is the affine class $t \mapsto a + bt$.

Why calibration can improve efficiency. Suppose the calibrated score $m_n^\star$ converges to a population limit m_0, where

$$m_0(X) = g_0\{m(X)\}, \qquad g_0 \in \arg\min_{g \in \mathcal{H}} \mathbb{E}_{P_0}\big[\{Y - g(m(X))\}^2\big].$$

Thus, m_0 is the best predictor obtainable from the original score m using post-processing maps in $\mathcal{H}$. When $m_n^\star$ is close to m_0, $\widehat{\psi}_{\text{cal}}$ behaves like the AIPW estimator that uses m_0. Proposition 5 therefore implies that this limiting estimator has asymptotic variance no larger than that of the AIPW estimator based on any $\theta \circ m$ with $\theta \in \mathcal{H}$. In particular, if $\mathcal{H}$ contains the identity map, then the estimator based on m_0 is asymptotically at least as efficient as the AIPW estimator based on the original score m.

3.2 Isotonic-calibrated PPI

We now turn to isotonic regression, the main nonparametric second-stage calibrator. Fitted on the labeled pairs $\{(m(X_i), Y_i)\}_{i=1}^n$, it is tuning-free, easy to compute, and cannot worsen empirical squared error relative to the original score [127, 70, 6].

The isotonic-calibrated plug-in estimator is

$$\widehat{\psi}_{\text{iso}} := \frac{1}{n+N}\left\{\sum_{i=1}^n m_{n,\text{iso}}^\star(X_i) + \sum_{j=1}^N m_{n,\text{iso}}^\star(\widetilde{X}_j)\right\},$$

where

$$m_{n,\text{iso}}^\star(X) = \hat{f}\{m(X)\}, \qquad \hat{f} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \big\{Y_i - f(m(X_i))\big\}^2,$$

and $\mathcal{F}_{\text{iso}}$ denotes the class of nondecreasing real-valued functions. The fitted values $m^\star_{n,\text{iso}}(X_i) = \hat{f}(m(X_i))$ are piecewise constant and can be computed by the pooled-adjacent-violators algorithm. This structure implies full empirical calibration even though the fitting class is isotonic.

Proposition 1 (Isotonic calibration is fully calibrated)**.** *For every function $h : \mathbb{R} \to \mathbb{R}$,*

$$\mathbb{P}_n^L\big[h(m^\star_{n,\text{iso}}(X))\{Y - m^\star_{n,\text{iso}}(X)\}\big] = 0. \tag{9}$$

Moreover, it holds that

$$\mathbb{P}_n^L\big[(Y - m^\star_{n,\text{iso}}(X))^2\big] \le \mathbb{P}_n^L\big[(Y - m(X))^2\big].$$

Thus isotonic regression is the main example in which a restricted monotone fitting class $\mathcal{H}$ yields a much larger calibration class $\mathcal{F}$. Consequently, Theorem 1 applies with $\mathcal{F}$ equal to the class of all real-valued functions. Additional implementation details appear in Appendices D and N.

3.3 Linearly calibrated PPI

For smaller labeled samples, one may prefer the lower-dimensional affine case. We therefore focus on linear calibration [65], where the fitting and calibration classes coincide as the affine maps. It regresses the outcome on the original score and corrects mean bias, scaling, and location shift within that class.

Let

$$\big(\hat{a}_{\text{lin}}, \hat{b}_{\text{lin}}\big) \in \arg\min_{a,b\in\mathbb{R}} \sum_{i=1}^n \{Y_i - a\, m(X_i) - b\}^2, \qquad m^\star_{n,\text{lin}}(X) := \hat{a}_{\text{lin}}\, m(X) + \hat{b}_{\text{lin}}. \tag{10}$$

Define the associated linearly calibrated plug-in estimator by

$$\widehat{\psi}_{\text{lin}} := \frac{1}{n+N}\left\{\sum_{i=1}^n m^\star_{n,\text{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\text{lin}}(\widetilde{X}_j)\right\}. \tag{11}$$

Because the affine class includes the constant function, the least-squares fit automatically enforces empirical mean calibration. As a result, the linearly calibrated plug-in estimator admits an exact AIPW representation.

Proposition 2 (Linear calibration as AIPW)**.** *The normal equations for equation (10) imply*

$$\mathbb{P}_n^L\{Y - m^\star_{n,\text{lin}}(X)\} = 0 \qquad \textit{and} \qquad \mathbb{P}_n^L\big[m^\star_{n,\text{lin}}(X)\{Y - m^\star_{n,\text{lin}}(X)\}\big] = 0.$$

Hence, for any $\hat{w}(X) = a + b\, m^\star_{n,\text{lin}}(X)$,

$$\widehat{\psi}_{\text{lin}} = \rho_n \mathbb{P}_n^L\{m^\star_{n,\text{lin}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^\star_{n,\text{lin}}(\widetilde{X})\} + \mathbb{P}_n^L\big[\hat{w}(X)\{Y - m^\star_{n,\text{lin}}(X)\}\big]. \tag{12}$$

The next result connects linear calibration to PPI++, a closely related estimator that chooses a rescaling of the original score by empirical efficiency maximization. In the mean-estimation problem, these two approaches are first-order equivalent, although they need not coincide exactly in finite samples. Intuitively, both are trying to make the same one-dimensional adjustment: they choose the best linear rescaling of $m(X)$. Let $\widehat{\psi}_{++}(\widehat{\lambda}_{++})$ denote the PPI++ estimator of Angelopoulos et al. [2], that is, the AIPW estimator based on the rescaled score $\widehat{\lambda}_{++} m(X)$, where $\widehat{\lambda}_{++}$ is obtained by empirical efficiency maximization over $\{\lambda m(X) : \lambda \in \mathbb{R}\}$ [90, 114]. The key point is that $\widehat{\lambda}_{++}$ estimates λ_0 in (2), which is exactly the population slope in the linear regression of Y on $m(X)$.

Proposition 3 (PPI++ and Linear Calibration)**.** *Assume that* $\mathrm{Var}\{m(X)\} > 0$, $\mathbb{E}_{P_0}[Y^2] < \infty$, *and* $\mathbb{E}_{P_0}[m(X)^2] < \infty$. *Then*

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = (1-\rho_n)\big(\hat{\lambda}_{++} - \hat{a}_{\text{lin}}\big)\left\{\mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X)\right\}.$$

Moreover, $\hat{\lambda}_{++} - \hat{a}_{\text{lin}} = O_p(n^{-1/2})$, *and, under* $\rho_n \to \rho_0 \in (0,1)$,

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = O_p\left(\frac{1-\rho_n}{\sqrt{\rho_n}}\,(n+N)^{-1}\right) = O_p\big((n+N)^{-1}\big).$$

In particular,

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = o_p\big(M^{-1/2}\big).$$

Linear calibration can thus be viewed as classical prognostic-score adjustment in semisupervised form [40, 84, 64, 96]. Richer smooth and generalized-linear-model extensions, including Platt scaling, Venn–Abers shrinkage, and cross-fitted calibration, are discussed in Appendix D; see in particular Appendix D.1.

4 Theory for isotonic-calibrated PPI

We now study the theoretical basis for *calibeating* via isotonic calibration. We establish asymptotic linearity, asymptotic normality, valid Wald inference, and efficiency results, and then show that no further post-processing of the fitted isotonic score yields additional first-order gain. To keep the exposition light, we focus on the non-cross-fitted estimator and treat the initial black-box score m as fixed.

4.1 Asymptotic normality, inference, and efficiency

Throughout this subsection, let f_0 denote a population isotonic calibration map for the fixed score m, defined by

$$f_0 \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \mathbb{E}_{P_0}\big[\{Y - f(m(X))\}^2\big], \qquad m_0 = f_0 \circ m. \tag{13}$$

Thus, m_0 is the population L^2-optimal monotone transformation of the initial score $m(X)$.

Assumption 2 (Regularity conditions for isotonic calibration)**.** Assume:

(i) **Boundedness:** There exists $C_0 < \infty$ such that $|\mathbb{E}_{P_0}[Y \mid m(X)]| \le C_0$ almost surely and $\sup_x |m^\star_{n,\text{iso}}(x)| = O_p(1)$.

(ii) **Controlled tails:** The conditional error $Y - \mathbb{E}_{P_0}[Y \mid m(X)]$ is sub-Gaussian or subexponential under P_0.

Under Assumption 2, the isotonic estimator attains the classical rate $\|m^\star_{n,\text{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3})$; see Chatterjee et al. [14], Yang and Barber [125].

Theorem 2 (Asymptotic linearity under the two-sample model)**.** *Suppose Assumptions 1 and 2 hold. Then*

$$\begin{aligned}\widehat{\psi}_{\text{iso}} - \psi_0 = \rho_n(\mathbb{P}^L_n - P_0)\Big\{m_0 - \psi_0 + \rho_n^{-1}(Y - m_0)\Big\} \\ + (1-\rho_n)(\mathbb{P}^U_N - P_{0,X})\{m_0 - \psi_0\} + O_p(n^{-2/3}) + O_p\big(n^{-1/3}N^{-1/2}\big).\end{aligned} \tag{14}$$

Thus, under the $\sqrt{M}$ scaling, $\widehat{\psi}_{\text{iso}}$ is asymptotically linear with influence pair $(D^L_{m_0}, D^U_{m_0})$ given by

$$D^L_{m_0}(X,Y) = m_0(X) - \psi_0 + \rho_0^{-1}\{Y - m_0(X)\}, \qquad D^U_{m_0}(\widetilde{X}) = m_0(\widetilde{X}) - \psi_0.$$

Corollary 1 (Asymptotic normality and Wald inference)**.** *Under the conditions of theorem 2, assume that*

$$\sigma_0^2 := \rho_0 \operatorname{Var}\{D^L_{m_0}(X,Y)\} + (1-\rho_0)\operatorname{Var}\{D^U_{m_0}(\widetilde{X})\} > 0.$$

Define the empirical variance

$$\widehat{\sigma}^2 := \rho_n \mathbb{P}^L_n[(\widehat{D}^L)^2] + (1-\rho_n)\mathbb{P}^U_N[(\widehat{D}^U)^2],$$

where

$$\widehat{D}^L_i := m^\star_{n,\text{iso}}(X_i) - \widehat{\psi}_{\text{iso}} + \rho_n^{-1}\{Y_i - m^\star_{n,\text{iso}}(X_i)\}, \qquad \widehat{D}^U_j := m^\star_{n,\text{iso}}(\widetilde{X}_j) - \widehat{\psi}_{\text{iso}}.$$

Then

$$\frac{\sqrt{M}\,(\widehat{\psi}_{\text{iso}} - \psi_0)}{\widehat{\sigma}} \xrightarrow{d} N(0,1).$$

Corollary 2 (Efficiency regimes)**.** *Under the conditions of theorem 2, the following hold.*

(i) If $m_0 = \mu_0$ almost surely, then $\widehat{\psi}_{\text{iso}}$ is efficient for ψ_0 in the full two-sample experiment.

(ii) If $\mathbb{E}_{P_0}[Y \mid m(X)]$ is monotone increasing in $m(X)$, then $m_0(X) = \mathbb{E}_{P_0}[Y \mid m(X)]$ almost surely. Hence $\widehat{\psi}_{\text{iso}}$ is efficient for ψ_0 in the reduced two-sample experiment obtained by replacing X with the scalar score $m(X)$.

4.2 No further first-order gain from post-processing

We now ask whether the isotonic-calibrated score can be improved by an additional post-processing step. The key observation is that $\widehat{\psi}_{\text{iso}}$ uses the covariates only through the one-dimensional score $m^{\star}_{n,\text{iso}}(X)$. It is therefore natural to view the estimator as operating in the reduced two-sample experiment

$$\mathcal{E}_{m^{\star}_{n,\text{iso}}} := \Big(\{(m^{\star}_{n,\text{iso}}(X_i), Y_i)\}_{i=1}^{n},\ \{m^{\star}_{n,\text{iso}}(\widetilde{X}_j)\}_{j=1}^{N}\Big),$$

in which the full covariate X is replaced by the scalar score $m^{\star}_{n,\text{iso}}(X)$.

Assumption 3 (Bounded-variation score regression)**.** The map $t \mapsto \mathbb{E}_{P_0}[Y \mid m(X) = t]$ has bounded variation on the support of $m(X)$.

Proposition 4 (No further first-order improvement from post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\widehat{\psi}_{\text{eff}} := \widehat{\psi}(\widehat{m}_0), \qquad \widehat{m}_0(X) := \mathbb{E}_{P_0}[Y \mid m^{\star}_{n,\text{iso}}(X)]$$

denote the benchmark estimator in the reduced experiment based on the isotonic score $m^{\star}_{n,\text{iso}}(X)$. Then

$$\|\widehat{m}_0 - m^{\star}_{n,\text{iso}}\|^2_{2,P_{0,X}} = O_p(n^{-2/3}),$$

and

$$\widehat{\psi}_{\text{eff}} - \widehat{\psi}_{\text{iso}} = o_p(M^{-1/2}).$$

In particular, $\widehat{\psi}_{\text{iso}}$ and $\widehat{\psi}_{\text{eff}}$ have the same asymptotic variance.

5 Experiments

Across all experiments we use the same comparison hierarchy. The labeled-only estimator (reported as NAIVE in the simulation) is the safety baseline. AIPW is the main raw-score baseline, with PPI as the weaker comparator. PPI++ and AIPW-EM are the efficiency-rescaling competitors, and `LinearCal`, `AutoCal`, and related calibration rules are the proposed fix when the score is miscalibrated. The three subsections below therefore move from a controlled mechanism check, to reproduced benchmark evidence, to a practical LLM-evaluation setting.

5.1 Simulation study

We first study a controlled binary-outcome simulation designed to isolate score miscalibration. Let $S \sim N(0,1)$, define

$$\mu_0(S) = \{1 + \exp(-5S)\}^{-1},$$

and draw $Y \mid S \sim \text{Bernoulli}\{\mu_0(S)\}$. To induce substantial miscalibration, we apply an affine-biased monotone distortion on the logit scale,

$$m(X) = \text{clip}\left[-0.15 + 0.75\,\sigma\big\{0.8\,z(S) + 0.1\,z(S)^3 - 1\big\},\ 0.01,\ 0.99\right],$$

where $z(S) := \text{logit}\{\mu_0(S)\}$. We compare NAIVE, PPI, AIPW, clipped PPI++, unclipped AIPW-EM, and the calibration-based estimators LINEARCAL, MONOSPLINE, ISOCAL, and AUTOCAL. The labeled sample size ranges over $n \in \{50, 100, 200, 400, 800, 1200, 2400\}$, with $N/n \in \{1, 16\}$, and each setting uses 500 repetitions.

Figure 2 isolates the mechanism. All score-based AIPW-type estimators are essentially unbiased here, so the comparison is mainly about efficiency. AIPW consistently improves on PPI among the raw-score methods. When the unlabeled sample is large, unclipped AIPW-EM remains close to `LinearCal`, whereas clipped PPI++ is pulled back toward PPI once the variance-minimizing rescaling exceeds 1. Under genuine miscalibration, the gains therefore come from calibration, with `AutoCal` staying near the best fixed rule.

5.2 Empirical Illustration

We next evaluate the same estimators on the original mean-estimation benchmarks of Angelopoulos et al. [1, 2], as provided in the `ppi_py` package. Using the same datasets and labeled-sample-size grids, we repeatedly split each fully labeled benchmark into labeled and unlabeled subsets, evaluate

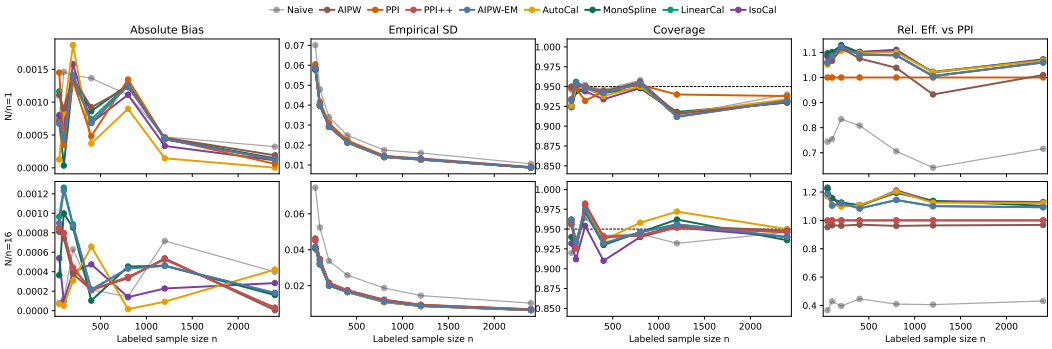

Figure 2: Binary simulation with a miscalibrated score. Columns report bias, standard deviation, coverage, and relative efficiency versus PPI for $N/n \in \{1, 16\}$.

all estimators on the same split, treat the full-sample mean as ground truth, and average over 200 random splits. We report `forest`, `galaxies`, and `census_income`. The main-text figure focuses on PPI, AIPW, PPI++, AIPW-EM, `AutoCal`, and `LinearCal`; Appendix M gives the richer calibration comparison.

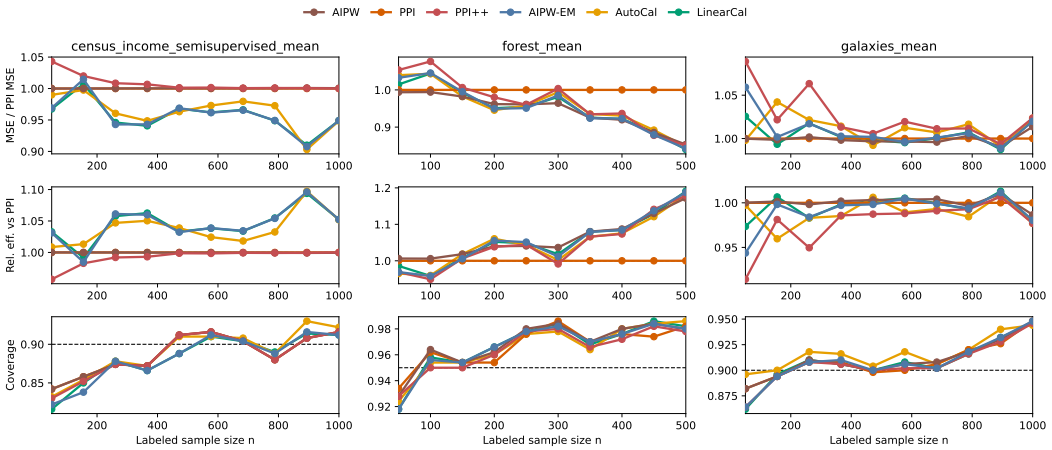

Figure 3: Benchmark summary for the reproduced `ppi_py` datasets. Panels report normalized MSE, relative efficiency, and coverage.

Figure 3 shows the same pattern on the reproduced benchmarks. AIPW is the stronger raw-score baseline, with PPI competitive only when the unlabeled sample is extremely large. The official `ppi_py` implementation clips the PPI++ scaling coefficient to $[0, 1]$, so when the variance-minimizing rescaling exceeds 1, clipped PPI++ is pulled back toward PPI whereas unclipped AIPW-EM tracks `LinearCal` much more closely. Across the 30 dataset-size cells in Figure 3, AIPW-EM has lower MSE than clipped PPI++ in 29 and stays nearly tied with `LinearCal`. Post-hoc calibration is most useful when the score is meaningfully miscalibrated and the labeled sample is large enough to fit the second-stage calibrator stably, with `LinearCal` strongest on `census_income` and `AutoCal` close to the best fixed option.

5.3 LLM evaluation benchmark

We now turn to a practical LLM-evaluation setting where the same comparison hierarchy is applied to real public model outputs, public evaluator scores, and public labels. The first track uses PPE Human Preference V1, where the estimand for each target model is its human win share across prompt–response comparisons, with ties counted as half-wins. The second uses a PPE correctness suite over MMLU-Pro, MATH, GPQA, IFEval, and MBPP+, where each unit compares a sampled conflict pair from the same generator model and the estimand is the correctness rate of the first

answer in the pair. In both tracks the prompts, outputs, judgments, and evaluator scores are real; only the labeled/unlabeled split is rerandomized. Across 200 repeated splits we compare labeled-only estimation against raw PPI/AIPW, the efficiency-maximized variants, and the calibration-based estimators from our main experiments. The main-text summary in Figure 4 and table 1 therefore focuses on the completed PPE tracks only, while we leave RewardBench to a separate robustness analysis.

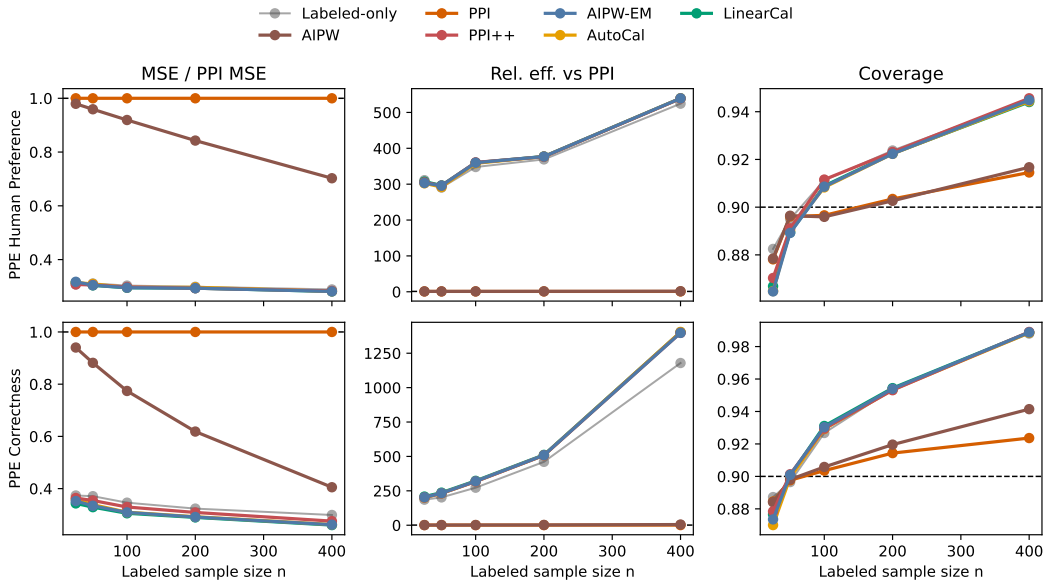

Figure 4: PPE-centered LLM-evaluation benchmark. Panels report normalized MSE relative to PPI, relative efficiency versus labeled-only, and coverage after macro-averaging across the public PPE evaluator models. We omit RewardBench from the main-text figure until its separate robustness run completes cleanly.

n	Estimator	Human MSE	Human label savings	Correctness MSE	Correctness label savings	Avg. cov.
100	Labeled-only	0.001425	0.0%	0.002010	0.0%	0.919
100	PPI	0.495213	0.4%	0.533854	1.0%	0.900
100	AIPW	0.440144	0.4%	0.352885	0.9%	0.901
100	PPI++	0.001395	2.7%	0.001875	7.0%	0.921
100	AIPW-EM	0.001392	3.1%	0.001826	9.5%	0.919
100	LinearCal	0.001391	3.1%	0.001813	10.1%	0.920
100	AutoCal	0.001405	2.8%	0.001835	9.4%	0.919
100	MonoSpline	0.001412	2.5%	0.001839	9.2%	0.915
400	Labeled-only	0.000286	0.0%	0.000248	0.0%	0.966
400	PPI	0.147604	0.5%	0.272679	2.0%	0.919
400	AIPW	0.087701	0.4%	0.042829	0.9%	0.929
400	PPI++	0.000279	2.5%	0.000229	8.3%	0.967
400	AIPW-EM	0.000278	2.8%	0.000223	10.3%	0.967
400	LinearCal	0.000278	2.8%	0.000222	10.5%	0.966
400	AutoCal	0.000279	2.7%	0.000223	10.2%	0.966
400	MonoSpline	0.000279	2.7%	0.000227	9.4%	0.965

Table 1: PPE-only numerical summary at $n \in \{100, 400\}$. Human and correctness columns report MSE and implied label savings relative to labeled-only; Avg. cov. averages coverage across the two PPE tracks.

Two messages stand out. First, raw proxy use can fail badly. Raw AIPW is again the stronger raw-score baseline, but even AIPW can be dramatically worse than labeled-only when the reward-model margin is badly scaled. On PPE Human at $n = 100$, raw PPI and raw AIPW have MSE 0.495 and 0.440, compared with 0.001425 for labeled-only and 0.001391 for `LinearCal`; on PPE Correctness at $n = 100$, the corresponding MSEs are 0.534, 0.353, 0.002010, and 0.001813. These

estimators remain valid first-order corrections, but the extra proxy term can inflate variance enough that labeled-only is far safer.

Second, calibration makes the same evaluator scores practically useful. `LinearCal` is the strongest and most stable method overall, with `AutoCal` close behind. On PPE Human, `LinearCal` is best from $n = 50$ onward, improving on labeled-only from 0.001425 to 0.001391 at $n = 100$ and from 0.000286 to 0.000278 at $n = 400$, corresponding to modest 2.5% to 3.1% label savings. On PPE Correctness, the gains are clearer: `LinearCal` is best at every budget, improving from 0.002010 to 0.001813 at $n = 100$ and from 0.000248 to 0.000222 at $n = 400$, with about 7.5% to 10.5% label savings. `MonoSpline` substantially improves over raw PPI/AIPW and is usually competitive with PPI++ and AIPW-EM, but it is slightly less stable than `LinearCal` and usually a bit worse than `AutoCal`.

Third, the evaluator-specific appendix breakdown (Figure 7) shows that the most extreme raw-proxy failures are driven by evaluators such as Skywork and Athene, whereas ArmoRM is much more benign. Finally, calibration also helps on the downstream ranking task induced by PPE Human. Raw PPI/AIPW recover the model ordering poorly, whereas calibrated and efficiency-maximized methods recover it well. At $n = 200$, `AutoCal` attains the highest average Spearman correlation, about 0.905; at $n = 400$, `LinearCal` gives the best overall ranking quality, with Spearman about 0.941 and top-1 identification about 0.646. We therefore view calibration not as a marginal efficiency tweak, but as the step that makes cheap public LLM evaluators safe enough to use in practice; see Appendix M.2.

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A Extended related work

Although PPI has been presented as a recent development, the underlying problem and many of its estimators are special cases of a broader literature on semiparametric debiasing and regression adjustment, including work on missing data, causal inference, and debiased machine learning.

Semiparametric statistics and missing data. This paper builds on foundational ideas from the missing-data and semiparametric literatures. Rubin [91] formalized missing at random as a key identification condition for recovering full-data targets from partially observed outcomes. Semiparametric efficiency theory then characterized efficient influence functions, regular asymptotically linear estimators, and one-step debiased procedures [60, 74, 41, 54, 10]. In missing-data and causal-inference problems, these ideas led to influence-function-based estimating equations [81, 82, 80, 114, 101], often with doubly robust structure [5]. As a result, the efficiency theory for missing-data models under missingness or coarsening at random is now well understood, and the class of regular asymptotically linear estimators in these models can be characterized through their influence functions [81, 82, 80, 114, 101].

Debiased machine learning and targeted learning. A large body of work combines the semiparametric framework above with machine-learning-based nuisance estimation, often under the label of debiased machine learning. Flexible learning methods within estimating equations and AIPW-style procedures already appear in van der Laan and Robins [114]. Targeted minimum loss estimation (TMLE), or targeted learning, updates an initial nuisance estimator in a parameter-targeted way so that the resulting plug-in estimator is debiased; in the present setting, the main nuisance is the outcome regression [116, 115, 33, 44, 85]. These approaches are often combined with sample splitting and cross-fitting to accommodate generic machine-learning estimators [95, 54, 129, 115, 22]. Double machine learning emphasizes similar ingredients—orthogonality, cross-fitting, and influence-function expansions—for inference with modern nuisance estimators [22, 24], and can be understood within the same semiparametric framework [111, 28, 20]. For sufficiently stable estimators, cross-fitting can itself be relaxed [17]. Closely related work studies balancing-weight estimators [38, 48, 131, 58, 16, 42], which often admit equivalent regression-adjustment or augmented representations [15, 11, 89]. In completely missing-at-random settings, including semisupervised learning and randomized trials, these approaches allow flexible black-box nuisance estimation while still supporting valid inference under appropriate conditions [66, 67, 84, 46, 45].

Semisupervised inference. Related work studies semisupervised or surrogate-assisted estimation in settings where the primary outcome is observed only on a limited subset, while auxiliary outcomes

or surrogate measurements are available more broadly [72, 73, 19, 18, 21, 49, 52]. Two-sample semisupervised inference is a special case of two-stage sampling designs [97, 83, 43, 78] and, more generally, of data fusion, for which a substantial semiparametric efficiency literature has been developed [61, 62, 32, 124]. More recent papers have used the term *prediction-powered inference* (PPI) for the semisupervised mean-estimation setting in which a black-box predictor is combined with a small labeled sample and a large unlabeled sample [1, 2, 130, 124, 99, 77, 57]. These papers also helped popularize such methods for a broader machine-learning audience and provided accessible software. Closely related questions have long been studied in randomized trials, where prognostic scores or other regression adjustments learned from larger historical or auxiliary data sources are used to improve precision in smaller trials [40, 84, 66–68, 96, 4, 46]. Methodologically, however, PPI estimators are best understood as instances or variants of existing semiparametric procedures. For example, PPI++ is an AIPW estimator [81, 114] obtained by empirical efficiency maximization [90] over a scaling parameter, and cross-PPI [130] is simply AIPW with cross-fitting. MEC [57] instead uses generalized entropy weights to balance the prediction score, which is closer to the balancing-weight representation of our linearly calibrated estimator. Our linearly calibrated estimator emphasizes the corresponding regression-adjustment view, implemented by ordinary linear regression of Y on the score. More fundamentally, the ability to combine black-box prediction models with valid inference is a consequence of the double robustness of AIPW when the missingness probability is known, and is not specific to semisupervised mean estimation [5, 98, 88].

From missing data to causal inference. From the perspective of estimation and efficiency theory, semisupervised mean estimation with missing-at-random outcomes is equivalent to randomized experiments, controlled trials, and survey-sampling designs once the missingness indicator is reinterpreted as a treatment or sampling indicator, so that the relevant potential outcomes are viewed as missing [92, 114, 30, 69]. This equivalence allows tools from covariate adjustment, causal inference, and model-assisted estimation to be transferred directly to the semisupervised setting; see, for example, Mozer [69] for results relating prediction-powered inference to model-assisted survey-sampling estimators. The two-sample formulation we study is also equivalent to a one-sample i.i.d. model [61]. In particular, our methods apply to inference on the counterfactual mean $\mathbb{E}[Y(1)]$ in a randomized trial with binary treatment, where treated units form the labeled sample and control units contribute covariate information only; see Appendix E.

Calibration. A final relevant thread concerns calibration. Classical post-processing methods such as Platt scaling [75, 25, 36], linear calibration [65, 23, 59], isotonic regression [127, 70], and histogram binning [126, 35] improve predictive reliability without retraining the underlying model. The idea has deeper roots in the forecasting literature [65, 63, 121, 120, 31, 55]; see also the historical discussion in Lee et al. [56]. More recent work studies calibration under broader losses and prediction tasks [51, 71, 106, 123, 102], as well as applications to conformal prediction [103, 102], importance-weight stabilization [37, 27, 3, 107], treatment effect estimation [23, 106, 123], and value prediction in offline reinforcement learning [104]. Distribution-free guarantees for histogram binning are developed by Gupta and Ramdas [35], while van der Laan et al. [106], van der Laan and Alaa [102] establish asymptotic distribution-free guarantees for isotonic calibration. Most closely related to our work is recent research connecting calibration to debiased estimation of statistical functionals [108, 105, 110]. In particular, van der Laan et al. [108] studies calibrated debiased machine learning for doubly robust inference, which in our setting reduces to calibrating the outcome regression and exploiting the resulting plug-in representation, while van der Laan et al. [110] studies calibrated plug-in regression estimators in reinforcement learning. More broadly, van der Laan et al. [105] shows how plug-in calibration can support regression and treatment effect estimation, including superefficient inference through dimension reduction.

B Semiparametric background and equivalence material

B.1 Review of AIPW, PPI, and semiparametric efficiency

The AIPW class. We begin with the augmented inverse-probability weighted (AIPW) estimator in the unrestricted two-sample model [81]. It is helpful to view AIPW not as a single estimator, but as a family of estimators indexed by a score function $f : \mathcal{X} \to \mathbb{R}$ with finite second moment. For any such f, define

$$\widehat{\psi}(f) := \rho_n \mathbb{P}_n^L\{f(X)\} + (1-\rho_n)\mathbb{P}_N^U\{f(\widetilde{X})\} + \mathbb{P}_n^L\{Y - f(X)\}. \tag{15}$$

This estimator combines a plug-in term that imputes Y from $f(X)$ with a residual correction from the labeled sample. For any fixed f, $\widehat{\psi}(f)$ is unbiased in finite samples for $\psi_0 = \mathbb{E}_{P_0}[Y]$. In particular, $f(X) = 0$ yields the empirical mean estimator $\mathbb{P}_n^L\{Y\}$. Thus, the choice of f affects efficiency, through the estimator's variance, but not validity, in the sense that the resulting confidence intervals remain valid. This makes AIPW a "safe" estimator [124]. This property can be viewed as a particularly simple instance of double robustness[1] [5]. The same basic principle underlies the use of flexible machine learning for covariate adjustment in randomized trials while retaining valid inference [66, 67, 96, 46].

Asymptotic linearity and efficiency. To characterize the leading large-sample error in the two-sample model, we decompose the estimation error into one contribution from the labeled sample and one from the unlabeled sample. We say that $\widehat{\psi}$ is asymptotically linear with influence pair (D^L, D^U) if

$$\widehat{\psi} - \psi_0 = \frac{1}{M}\sum_{i=1}^{n} D^L(X_i, Y_i) + \frac{1}{M}\sum_{j=1}^{N} D^U(\widetilde{X}_j) + o_p(M^{-1/2}),$$

where $\mathbb{E}_{P_0}[D^L(X,Y)] = 0$ and $\mathbb{E}_{P_{0,X}}[D^U(\widetilde{X})] = 0$. By the central limit theorem and the independence of the two samples,

$$\sqrt{M}\,(\widehat{\psi} - \psi_0) \rightsquigarrow N\Big(0,\ \rho_0 \mathrm{Var}_{P_0}\{D^L(X,Y)\} + (1-\rho_0)\mathrm{Var}_{P_{0,X}}\{D^U(\widetilde{X})\}\Big).$$

The influence pair therefore determines the estimator's asymptotic variance and provides a convenient way to compare efficiency across different choices of f. The next result uses this representation to characterize the AIPW class, identify the efficient influence pair in the unrestricted two-sample model, and thereby obtain the semiparametrically efficient variance bound [10].

Proposition 5 (Influence pair for the AIPW class)**.** *Under Assumption 1, for any fixed $f : \mathcal{X} \to \mathbb{R}$ with finite second moment, the estimator $\widehat{\psi}(f)$ is asymptotically linear with influence pair (D_f^L, D_f^U), where*

$$D_f^L(X,Y) = \tilde{f}(X) - \psi_0 + \rho_0^{-1}\{Y - \tilde{f}(X)\}, \tag{16}$$

$$D_f^U(\widetilde{X}) = \tilde{f}(\widetilde{X}) - \psi_0, \tag{17}$$

with $\tilde{f}(X) := f(X) - \mathbb{E}_{P_{0,X}}\{f(X)\} + \psi_0$ denoting the recentered version of $f(X)$. The resulting asymptotic variance of $\sqrt{M}\,(\widehat{\psi}(f) - \psi_0)$ is

$$\mathrm{Var}\{\mu_0(X)\} + \rho_0^{-1}\mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big] + \frac{1-\rho_0}{\rho_0}\mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big].$$

This variance is minimized over the AIPW class when $\tilde{f} = \mu_0$, in particular at $f = \mu_0$. Therefore, the efficient influence pair for $\psi_0 = \mathbb{E}_{P_0}[Y]$ in the unrestricted two-sample model is $(D_{\mu_0}^L, D_{\mu_0}^U)$, and the corresponding asymptotic variance is the efficiency bound[2].

A key insight of Proposition 5 is that the limiting variance of the AIPW estimator depends on the score function f only through the mean squared error of its recentered version $\tilde{f}$, namely $\mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big]$. Thus, within the AIPW class, efficiency is maximized by choosing f so that $\tilde{f}$ approximates $\mu_0(X)$ as well as possible, with equality when $\tilde{f} = \mu_0$, in particular at $f = \mu_0$. This shows that full efficiency generally requires using the full covariate information X, or at least a summary rich enough to recover $\mu_0(X)$ [80]. In practice, this motivates taking f to be an estimated prediction score $m(X)$, learned using flexible black-box methods.

[1] In more general missing-at-random settings with unknown, covariate-dependent missingness probabilities, the analogous AIPW estimator remains doubly robust asymptotically, but it no longer enjoys the same finite-sample unbiasedness as in the present known-ρ_n setting [98].

[2] Appendix B.2 shows that every regular estimator is asymptotically equivalent to $\widehat{\psi}(f)$ for some score function f. Thus, the AIPW family characterizes the full class of regular estimators and the semiparametric efficiency bound in this model.

AIPW, PPI, and empirical-efficiency maximization. We now record several familiar choices of f within the class (15). The most common choice is $f(X) = m(X)$, where $m(X)$ is a prediction score, yielding the usual augmented inverse-probability weighted (AIPW) estimator [81]:

$$\widehat{\psi}_{\mathrm{AIPW}} = \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}.$$

Equivalently,

$$\widehat{\psi}_{\mathrm{AIPW}} = \frac{1}{M}\left\{\sum_{i=1}^{n} m(X_i) + \sum_{j=1}^{N} m(\widetilde{X}_j) + \frac{1}{\rho_n}\sum_{i=1}^{n}\{Y_i - m(X_i)\}\right\},$$

where ρ_n^{-1} is the inverse probability of observing the outcome.

Another estimator in this class was considered by Angelopoulos et al. [1] under the name PPI,[3] and we adopt that terminology here. It corresponds to the score choice

$$f(X) = \frac{m(X)}{1-\rho_n}.$$

Substituting this choice into (15) gives

$$\widehat{\psi}_{\mathrm{PPI}} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}.$$

Relative to $\widehat{\psi}_{\mathrm{AIPW}}$, $\widehat{\psi}_{\mathrm{PPI}}$ uses only the unlabeled sample in the plug-in term and rescales the score by the factor $(1-\rho_n)^{-1} > 1$. By Proposition 5, this helps only if the rescaled score is closer to $\mu_0(X)$ in mean squared error. Thus, PPI is inefficient when $m = \mu_0$, and is typically less efficient when m is already reasonably accurate. When ρ_n is small and m is sufficiently inaccurate, however, neither estimator is necessarily more efficient than the other [39, 87].

Because both $\widehat{\psi}_{\mathrm{AIPW}}$ and $\widehat{\psi}_{\mathrm{PPI}}$ use scores in the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, it is natural to choose the scaling factor data-adaptively rather than fixing it in advance. One way to do so is to minimize the estimator's empirical influence-function variance over this class:

$$\sigma_n^2(f) := \rho_n \mathbb{P}_n^L\left[\left\{f(X) - \widehat{\psi}(f) + \rho_n^{-1}(Y - f(X))\right\}^2\right] + (1-\rho_n)\mathbb{P}_N^U\left[\left\{f(\widetilde{X}) - \widehat{\psi}(f)\right\}^2\right].$$

That is, one chooses λ to minimize $\sigma_n^2(f)$ over $f(X) = \lambda m(X)$, a procedure known as empirical-efficiency maximization [90, 114]. The resulting estimator, called PPI++ by Angelopoulos et al. [2], is simply the AIPW estimator with an empirically rescaled prediction rule $\hat{\lambda} m(X)$, where $\hat{\lambda}$ estimates the population-optimal scaling coefficient

$$\lambda^\star = \frac{\mathrm{Cov}\big(Y, m(X)\big)}{\mathrm{Var}\big(m(X)\big)}. \tag{18}$$

The same principle extends beyond simple rescaling. For example, one can minimize $\sigma_n^2(f)$ over richer classes such as $\{\lambda m(X) + \beta_{\mathrm{lin}}^\top X : \lambda \in \mathbb{R},\ \beta_{\mathrm{lin}} \in \mathbb{R}^d\}$, over monotone transformations of the score, or over a reproducing kernel Hilbert space (Appendix D). More generally, the same criterion can be used in cross-validation to choose among candidate estimator classes [90].

B.2 Additional background on semiparametric statistics

The next result shows that the efficient choice in Proposition 5 is not merely optimal within the AIPW class. In fact, every regular estimator[4] in the unrestricted two-sample model is asymptotically equivalent to $\widehat{\psi}(f)$ for some score function f. Thus, the AIPW family characterizes the full class of regular estimators at first order in this model.

[3] The PPI and PPI++ estimators are algebraically equivalent to the model-assisted survey-design estimators of Cassel et al. [13] and Särndal et al. [93], respectively; see Mozer [69] for a review.

[4] Informally, a regular estimator is one whose large-sample behavior remains stable under small changes in the data-generating distribution [117].

Proposition 6 (Characterization of regular estimators)**.** *Under Assumption 1, a mean-zero pair (D^L, D^U) is the influence pair of a regular estimator of ψ_0 in the unrestricted two-sample model if and only if there exists a function $f_0 : \mathcal{X} \to \mathbb{R}$ with finite second moment, potentially depending on P_0, such that*

$$(D^L, D^U) = (D^L_{f_0}, D^U_{f_0}),$$

where D^L_f and D^U_f are given in (16)–(17). Equivalently, for every regular estimator $\widehat{\psi}^\dagger$, there exists such an f_0 for which

$$\widehat{\psi}^\dagger - \widehat{\psi}(f_0) = o_p(M^{-1/2}).$$

This characterization shows that the AIPW class is the relevant first-order benchmark in this model. The next remark illustrates that several estimators not always written in AIPW form—including regression adjustment, balancing-weight adjustment, and TMLE—can nonetheless be viewed as finite-sample or asymptotic equivalents of AIPW estimators [89, 116], while sometimes offering better finite-sample performance [76, 50, 109].

Remark 1 (Plug-in approaches to bias correction and TMLE)**. Regression plug-in estimators.** A particularly simple member of this first-order class is obtained through plug-in regression adjustment [94, 116, 84, 89]. Let

$$m_n(X) = \hat{a}_{\text{lin}}\, m(X) + \hat{b}_{\text{lin}}, \qquad (\hat{a}_{\text{lin}}, \hat{b}_{\text{lin}}) = \arg\min_{a,b} \sum_{i=1}^{n} \{Y_i - a\, m(X_i) - b\}^2.$$

The corresponding plug-in estimator is

$$\rho_n \mathbb{P}^L_n\{m_n(X)\} + (1-\rho_n)\mathbb{P}^U_N\{m_n(\widetilde{X})\}.$$

For any $\lambda \in \mathbb{R}$, this estimator also admits the AIPW representation

$$\rho_n \mathbb{P}^L_n\{m_n(X)\} + (1-\rho_n)\mathbb{P}^U_N\{m_n(\widetilde{X})\} + \lambda \mathbb{P}^L_n\big[Y - m_n(X)\big],$$

because the augmentation term $\mathbb{P}^L_n[Y - m_n(X)]$ is zero by the least-squares normal equations. Thus, when m_n converges to the best linear predictor of Y given $m(X)$, this plug-in estimator is typically at least as efficient as $\widehat{\psi}_{\text{AIPW}}$ and $\widehat{\psi}_{\text{PPI}}$, since it corresponds to an influence function with smaller or equal variance.

Balancing-weight estimators. The same estimator can also be written in terms of balancing weights. Let $w_n(X) = \hat{a}_{\text{bal}} + \hat{b}_{\text{bal}}\, m(X)$ be affine weights that balance $(1, m(X))$, in the sense that

$$\mathbb{P}^L_n\{w_n(X)\} = 1 \quad \text{and} \quad \mathbb{P}^L_n\big[m(X)w_n(X)\big] = \rho_n \mathbb{P}^L_n\{m(X)\} + (1-\rho_n)\mathbb{P}^U_N\{m(\widetilde{X})\}.$$

Then the corresponding balancing-weight estimator satisfies

$$\mathbb{P}^L_n\big\{w_n(X)Y\big\} = \rho_n \mathbb{P}^L_n\{m(X)\} + (1-\rho_n)\mathbb{P}^U_N\{m(\widetilde{X})\} + \mathbb{P}^L_n\big[w_n(X)\{Y - m(X)\}\big].$$

In fact, the plug-in and balancing-weight estimators are algebraically equivalent [15, 11]:

$$\mathbb{P}^L_n\big\{w_n(X)Y\big\} = \rho_n \mathbb{P}^L_n\{m_n(X)\} + (1-\rho_n)\mathbb{P}^U_N\{m_n(\widetilde{X})\}.$$

TMLE. More broadly, such debiased plug-in estimators can be viewed as instances of TMLE [116, 115, 45], a general framework for constructing plug-in estimators that are debiased and, in many cases, efficient by updating nuisance components to remove first-order bias. In the present setting, either the weights or the regression model are updated so that the corresponding augmentation term vanishes. In weighting-based TMLE, the update balances on the prediction model $m(X)$ (e.g., Hejazi and van der Laan 42); one may either include the known weight $1/\rho_n$ as an offset or balance on the intercept as well. In regression-based TMLE, the prediction model $m(X)$ may be included either as a covariate or as an offset (e.g., Lendle et al. 58), and the intercept adjustment accounts for the known inverse missingness-probability weight $1/\rho_n$. The specific targeting step is ultimately a practical choice, provided it solves the relevant estimating equation and removes the augmentation term; different choices may also yield additional benefits, such as further bias reduction or doubly robust inference [113, 12, 29, 8, 112, 108].

C Constructing confidence intervals via the bootstrap

Confidence intervals may be constructed using the standard Wald approximation from Corollary 1, namely

$$\widehat{\psi}_{\text{iso}} \pm z_{1-\alpha/2} \frac{\hat{\sigma}}{\sqrt{n+N}}.$$

As an alternative, one may use the nonparametric bootstrap and refit the isotonic calibration step within each bootstrap replicate, as proposed and theoretically studied by van der Laan et al. [108]; see also Tang and Westling [100]. This avoids direct estimation of the influence-function variance and automatically propagates uncertainty from estimation of the calibration map.

For each bootstrap replicate $b = 1, \ldots, B$, we sample with replacement from the labeled observations $\{(X_i, Y_i)\}_{i=1}^n$ and, independently, from the unlabeled observations $\{\widetilde{X}_j\}_{j=1}^N$, obtaining bootstrap samples

$$\{(X_i^{*(b)}, Y_i^{*(b)})\}_{i=1}^n, \qquad \{\widetilde{X}_j^{*(b)}\}_{j=1}^N.$$

We then refit the isotonic calibrator on the bootstrap labeled sample:

$$\hat{f}^{*(b)} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \{Y_i^{*(b)} - f(m(X_i^{*(b)}))\}^2,$$

and define

$$m_{\text{iso}}^{*(b)}(X) := \hat{f}^{*(b)}\{m(X)\}.$$

The corresponding bootstrap replicate of the estimator is

$$\widehat{\psi}_{\text{iso}}^{*(b)} := \frac{1}{n+N} \left\{ \sum_{i=1}^n m_{\text{iso}}^{*(b)}(X_i^{*(b)}) + \sum_{j=1}^N m_{\text{iso}}^{*(b)}(\widetilde{X}_j^{*(b)}) \right\}.$$

After computing B bootstrap replicates, one may form a percentile interval using the empirical $\alpha/2$ and $1-\alpha/2$ quantiles of

$$\{\widehat{\psi}_{\text{iso}}^{*(b)}\}_{b=1}^B.$$

Alternatively, one may use the bootstrap standard error

$$\widehat{\sigma}_{\text{boot}}^2 := \frac{1}{B-1} \sum_{b=1}^B \left(\widehat{\psi}_{\text{iso}}^{*(b)} - \bar{\psi}_{\text{iso}}^* \right)^2, \qquad \bar{\psi}_{\text{iso}}^* := \frac{1}{B} \sum_{b=1}^B \widehat{\psi}_{\text{iso}}^{*(b)},$$

and report the normal-approximation interval

$$\widehat{\psi}_{\text{iso}} \pm z_{1-\alpha/2}\, \widehat{\sigma}_{\text{boot}}.$$

The key implementation detail is that the isotonic regression step must be refit within each bootstrap replicate rather than held fixed. Otherwise, the resulting interval does not reflect uncertainty from estimation of the calibration map.

D Additional remarks

Remark 2 (Implementation)**.** Isotonic regression is widely available in standard software. A standard algorithm is the pooled-adjacent-violators algorithm (PAVA) [6]. Although it is generally fast, it may become slow at ultra-large sample sizes. A scalable alternative is to use regression tree software with monotonicity constraints, such as XGBoost or LightGBM (see Appendix N). An advantage of tree-based software is that the fit can be regularized by constraining or tuning the number of leaves, the maximum tree depth, or the minimum number of observations per leaf.

Remark 3 (Platt scaling adjustment)**.** The linear-calibration idea extends beyond squared-error loss. One can instead use logistic loss, Poisson loss, or, more generally, losses from canonical-link generalized linear models [84]. In particular, when $Y \in \{0, 1\}$ and $m(X) \in (0, 1)$ is an initial predictor,

one may apply Platt scaling [25, 75] by fitting a logistic regression of Y on the logit of $m(X)$. Writing $\eta_i(a,b) := a\,\mathrm{logit}(m(X_i)) + b$, this amounts to solving

$$\begin{aligned}\big(\hat{a}_{\mathrm{platt}}, \hat{b}_{\mathrm{platt}}\big) \in \arg\min_{a,b\in\mathbb{R}} \sum_{i=1}^{n} \Big[&-Y_i \log \sigma\big(\eta_i(a,b)\big) \\ &-(1-Y_i)\log\Big(1-\sigma\big(\eta_i(a,b)\big)\Big)\Big].\end{aligned} \tag{19}$$

with calibrated predictor

$$m^{\star}_{n,\mathrm{platt}}(X) := \sigma\big(\hat{a}_{\mathrm{platt}}\,\mathrm{logit}(m(X)) + \hat{b}_{\mathrm{platt}}\big),$$

where $\sigma(t) := (1+e^{-t})^{-1}$. Platt scaling may be preferred over linear regression calibration when predicted probabilities are systematically over- or under-confident, in which case the calibration curve is often approximately sigmoidal. More generally, outcomes can be rescaled to $[0,1]$ before applying this procedure. In online settings, calibration properties of Platt scaling are developed in [36].

D.1 Cross-Fitting and Cross-Calibration

In the main exposition, we assume for simplicity that the initial black-box predictor is fixed. In practice, however, it may be trained on the labeled sample itself, in which case the same labeled observations are used both to fit the predictor and to calibrate its scores. A cross-fitted version avoids this reuse by replacing the single prediction function with out-of-fold predictions and then fitting the calibrator to the resulting labeled out-of-fold scores.

1. Partition the labeled sample indices into folds $I_1, \ldots, I_K$.
2. For each fold k, train the black-box regression $m^{(-k)}$ on the labeled observations outside I_k, compute out-of-fold predictions $m^{(-k)}(X_i)$ for $i \in I_k$, and compute predictions $m^{(-k)}(\widetilde{X}_j)$ on the unlabeled sample.
3. Pool the labeled out-of-fold predictions $\{m^{(-k(i))}(X_i)\}_{i=1}^{n}$ and fit a calibrator f_n to the labeled pairs $(m^{(-k(i))}(X_i), Y_i)$.
4. Form calibrated predictions

$$m^{\star}_n(X_i) = f_n(m^{(-k(i))}(X_i))$$

and

$$m^{\star}_n(\widetilde{X}_j) = \frac{1}{K}\sum_{k=1}^{K} f_n(m^{(-k)}(\widetilde{X}_j)),$$

and then compute the plug-in estimator in equation (3).

This cross-fitted modification is the natural DML-style implementation when one wishes to reduce overfitting from using the same labeled data both to estimate the initial regression and to calibrate it for inference [129, 115, 22]. Related procedures have been used for debiased inference in van der Laan et al. [107, 108], Rabenseifner et al. [79], and a related cross-calibration variant appears in van der Laan et al. [106].

Remark 4 (Empirical efficiency maximization as a learning objective)**.** Following Rubin and van der Laan [90], the empirical variance criterion from Section B.1,

$$\sigma_n^2(f) := \rho_n \mathbb{P}_n^L\left[\Big\{f(X) - \widehat{\psi}(f) + \rho_n^{-1}(Y - f(X))\Big\}^2\right] + (1-\rho_n)\mathbb{P}_N^U\left[\Big\{f(\widetilde{X}) - \widehat{\psi}(f)\Big\}^2\right],$$

is not tied to the one-dimensional scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$. For any candidate class $\mathcal{G} \subset L^2(P_{0,X})$, one may instead define

$$\hat{f} \in \arg\min_{f\in\mathcal{G}} \sigma_n^2(f), \qquad \widehat{\psi}_{\mathrm{eem}} := \widehat{\psi}(\hat{f}),$$

so empirical efficiency maximization learns the augmentation $f(X)$ directly.

At the population level, let

$$V(f) := \rho_0 \mathrm{Var}\{D_f^L(X,Y)\} + (1-\rho_0)\mathrm{Var}\{D_f^U(\widetilde{X})\}$$

denote the asymptotic variance of $\widehat{\psi}(f)$. Using equations (16) and (17),

$$V(f) = \mathrm{Var}(Y) + \frac{1-\rho_0}{\rho_0}\mathrm{Var}\{Y - f(X)\}.$$

Equivalently, with $\tilde{f}(X) := f(X) - \mathbb{E}_{P_{0,X}}\{f(X)\} + \psi_0$,

$$V(f) = \mathrm{Var}(Y) + \frac{1-\rho_0}{\rho_0}\mathbb{E}\big[(Y - \tilde{f}(X))^2\big].$$

Thus the efficiency-optimal choice is the $L^2(P_{0,X})$-projection of μ_0 onto the normalized class $\{\tilde{f} : f \in \mathcal{G}\}$. If $\mathcal{G}$ is closed under additive constants, then minimizing $V(f)$ is equivalent to ordinary least-squares regression of Y on X over $\mathcal{G}$. In particular, rather than optimizing the empirical efficiency criterion $\sigma_n^2(f)$ directly, one can equivalently optimize the usual empirical mean-squared-error objective on the labeled sample.

This makes richer efficiency-maximizing classes immediate. For example, if $\mathcal{G} = \{b + h : h \in \mathcal{H},\ b \in \mathbb{R}\}$ for an RKHS $\mathcal{H}$ with kernel K, then a penalized version of empirical efficiency maximization is

$$(\hat{b}_\lambda, \hat{h}_\lambda) \in \arg\min_{b\in\mathbb{R},\ h\in\mathcal{H}} \left\{\mathbb{P}_n^L\big[(Y - b - h(X))^2\big] + \lambda\|h\|_{\mathcal{H}}^2\right\},$$

which is just kernel ridge regression on the labeled sample, up to the irrelevant additive constant in the variance criterion. By the representer theorem,

$$\hat{h}_\lambda(\cdot) = \sum_{i=1}^n \hat{\alpha}_i K(X_i, \cdot),$$

and, after centering the kernel or including the intercept explicitly,

$$\hat{\alpha} = (K_n + n\lambda I_n)^{-1}(Y - \hat{b}_\lambda \mathbf{1}_n).$$

Plugging $\hat{f}_\lambda = \hat{b}_\lambda + \hat{h}_\lambda$ into $\widehat{\psi}(\hat{f}_\lambda)$ yields the corresponding efficiency-maximized AIPW estimator.

For tuning and class selection, it is preferable to evaluate the same criterion out of fold. One may split the labeled sample into folds, fit f^{-k} on the training part of fold k, compute the held-out criterion $\sigma_{n,k}^2(f^{-k})$, and average over folds. The selected class or tuning parameter can then be used in a cross-fitted or cross-averaged AIPW estimator. This gives a direct cross-validated empirical-efficiency objective for choosing among classes such as linear, spline, isotonic, or RKHS-based regressors while preserving the usual sample-splitting protection against in-sample overfitting.

Remark 5 (Shrinkage via Venn–Abers)**.** When the labeled sample is small, isotonic calibration may be unstable. A natural alternative is Venn–Abers calibration, originally introduced for binary classification [119]; see [103, 102] for generalized versions beyond the binary setting. Assume $Y \in [0,1]$, or rescale otherwise. For each x, augment the labeled calibration sample with the hypothetical point $(m(x), y)$ for $y \in \{0,1\}$, fit the resulting isotonic calibrators $f_n^{(x,0)}$ and $f_n^{(x,1)}$, and form the interval-valued prediction

$$\big[f_n^{(x,0)}\{m(x)\},\ f_n^{(x,1)}\{m(x)\}\big].$$

Following [119], one may convert this interval to a point prediction by shrinking its midpoint toward the (unadjusted) AIPW estimator:

$$m_{n,\mathrm{VA}}^*(x) := m_{n,\mathrm{mid}}^*(x) + \big\{f_n^{(x,1)}\{m(x)\} - f_n^{(x,0)}\{m(x)\}\big\}\big\{\widehat{\psi}_{AIPW} - m_{n,\mathrm{mid}}^*(x)\big\},$$

where

$$m_{n,\mathrm{mid}}^*(x) := \frac{f_n^{(x,1)}\{m(x)\} + f_n^{(x,0)}\{m(x)\}}{2}.$$

Since the shrinkage magnitude is proportional to the width of the Venn–Abers interval, this construction can be more stable than plain isotonic calibration in small samples.

Remark 6 (Relation between linear calibration, PPI, and prognostic-score adjustment)**.** Linear calibration can be viewed as prognostic-score regression adjustment, a classical approach to improving precision through flexible covariate adjustment in small-sample settings [40, 84, 64, 96, 4, 46]. Proposition 3 shows that the PPI++ estimator admits the same first-order interpretation. Likewise, the PPI estimator $\widehat{\psi}_{\mathrm{PPI}}$ is algebraically equivalent to the unlabeled-sample plug-in estimator $\mathbb{P}_N^U\{m^\star_{n,\mathrm{const}}(X)\}$ based on the intercept-only adjustment

$$\hat{a}_{\mathrm{const}} \in \arg\min_{a\in\mathbb{R}} \sum_{i=1}^{n} \{Y_i - a - m(X_i)\}^2, \qquad m^\star_{n,\mathrm{const}}(X) := \hat{a}_{\mathrm{const}} + m(X), \tag{20}$$

which calibrates only the mean. By contrast, the standard AIPW estimator $\widehat{\psi}_{\mathrm{AIPW}}$ is equivalent to the pooled-sample mean $\rho_n \mathbb{P}_n^L\{m^\star_{n,\mathrm{const}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^\star_{n,\mathrm{const}}(X)\}$.

Remark 7 (Practical takeaway)**.** Isotonic calibration can improve efficiency when the original score is informative but miscalibrated. The resulting estimator behaves as if $m(X)$ were replaced by its best monotone calibration $m_0(X)$. Because the isotonic class contains the identity map and all affine maps with positive slope, the isotonic-calibrated estimator has limiting variance no larger than that of any estimator that is first-order equivalent to AIPW with score $\theta\{m(X)\}$ for such a transformation θ. This includes AIPW and PPI based on $m(X)$, as well as PPI++, and linear calibration whenever the corresponding scaling or slope coefficient is nonnegative. Thus isotonic calibration is asymptotically at least as efficient, at first order, as these methods, and can be strictly more efficient when the best monotone calibration is genuinely nonlinear. More generally, if $\mathbb{E}_{P_0}[Y \mid m(X)]$ is monotone, then no regular estimator based only on $m(X)$ can improve on the isotonic-calibrated estimator at first order.

Remark 8 (When full efficiency is attained)**.** Full efficiency requires that the calibrated target recover all regression information relevant to ψ_0. This holds if $m_0 = \mu_0$ almost surely. More generally, full efficiency fails when the one-dimensional score $m(X)$ omits predictive information in X that cannot be recovered by monotone calibration.

E Prediction-Powered Causal Inference via Calibration (i.e., Calibrated DML)

This appendix records the direct translation of the two-sample missing-outcome setup to randomized experiments. The central point is that the semisupervised problem studied in the main text and two-arm causal inference in a randomized trial are the same statistical problem once the unobserved potential outcome in one arm is interpreted as the missing outcome. We then illustrate how our calibrated plug-in estimators apply in this setting, where they arise as special cases of calibrated DML [108].

We use the standard potential-outcomes notation. Each unit has two potential outcomes, $Y(1)$ under treatment and $Y(0)$ under control, and the observed outcome satisfies

$$Y = AY(1) + (1-A)Y(0),$$

where $A \in \{0,1\}$ is the treatment assignment. The mean potential outcomes are

$$\mu_1 := \mathbb{E}\{Y(1)\}, \qquad \mu_0 := \mathbb{E}\{Y(0)\},$$

and the average treatment effect is $\tau := \mu_1 - \mu_0$. In a randomized trial, identification follows from the usual causal conditions: consistency and no interference, exchangeability induced by randomization,

$$A \perp (X, Y(1), Y(0)),$$

and positivity, so that each arm has positive assignment probability.

Now partition the observed data into the two arm-specific samples

$$\mathcal{D}_{n,1} := \{(X_i, Y_i) : A_i = 1\}, \qquad \mathcal{D}_{n,0} := \{(\widetilde{X}_j, \widetilde{Y}_j) : A_j = 0\},$$

with sample sizes $n_1 := |\mathcal{D}_{n,1}|$, $n_0 := |\mathcal{D}_{n,0}|$, and $n = n_1 + n_0$. Let $\mathbb{P}_{n,a}$ denote the empirical mean over arm a, and let $\hat{\pi}_a := n_a/n$. To estimate μ_1, one treats $\mathcal{D}_{n,1}$ as the labeled sample for $Y(1)$ and uses the covariates in $\mathcal{D}_{n,0}$ as the unlabeled sample. To estimate μ_0, one reverses the roles

of the two arms. Thus each mean potential outcome is a two-sample missing-outcome estimand of exactly the form analyzed in the main text.

For $a \in \{0,1\}$, let $m_a(X)$ be an arm-specific prediction score fit using only observations in arm a. The standard arm-specific AIPW estimator is

$$\hat{\mu}_a^{\mathrm{AIPW}} := \mathbb{P}_n\{m_a(X)\} + \mathbb{P}_n\left[\frac{\mathbf{1}(A=a)}{\hat{\pi}_a}\{Y - m_a(X)\}\right].$$

For $a \in \{0,1\}$, let $m^*_{n,a}(X)$ denote the calibrated version of the score $m_a(X)$. The calibrated estimator of μ_a is then

$$\hat{\mu}_a^{\mathrm{cal}} := \mathbb{P}_n\{m^*_{n,a}(X)\} = \hat{\pi}_a\,\mathbb{P}_{n,a}\{m^*_{n,a}(X)\} + (1-\hat{\pi}_a)\,\mathbb{P}_{n,1-a}\{m^*_{n,a}(X)\}. \tag{21}$$

This is exactly the pooled plug-in estimator from the two-sample formulation, with one arm contributing outcomes and the other contributing only covariates. The same estimator also has the exact AIPW form

$$\hat{\mu}_a^{\mathrm{cal}} = \mathbb{P}_n\{m^*_{n,a}(X)\} + \mathbb{P}_n\left[\frac{\mathbf{1}(A=a)}{\hat{\pi}_a}\{Y - m^*_{n,a}(X)\}\right]. \tag{22}$$

Because

$$\mathbb{P}_n\left[\frac{\mathbf{1}(A=a)}{\hat{\pi}_a}\{Y - m^*_{n,a}(X)\}\right] = \mathbb{P}_{n,a}\{Y - m^*_{n,a}(X)\},$$

the residual correction vanishes exactly whenever the calibrator includes the constant score and therefore enforces mean calibration within arm a. This is the randomized-trial analogue of the exact augmented representation in theorem 1.

The average treatment effect is then estimated by

$$\hat{\tau}^{\mathrm{cal}} := \hat{\mu}_1^{\mathrm{cal}} - \hat{\mu}_0^{\mathrm{cal}}. \tag{23}$$

Equivalently,

$$\hat{\tau}^{\mathrm{cal}} = \mathbb{P}_n\{\tilde{m}_1(X) - \tilde{m}_0(X)\} + \mathbb{P}_n\left[\frac{A}{\hat{\pi}_1}\{Y - \tilde{m}_1(X)\} - \frac{1-A}{\hat{\pi}_0}\{Y - \tilde{m}_0(X)\}\right].$$

Its influence function is the difference of the two arm-specific AIPW influences,

$$D_\tau(O) = \left\{\tilde{m}_1(X) - \mu_1 + \frac{A}{\pi_1}\big(Y - \tilde{m}_1(X)\big)\right\} - \left\{\tilde{m}_0(X) - \mu_0 + \frac{1-A}{\pi_0}\big(Y - \tilde{m}_0(X)\big)\right\}.$$

Accordingly, Wald inference for the ATE may be based on the empirical variance of the estimated influence values divided by n, or on the nonparametric bootstrap refitting the calibration step within each bootstrap sample.

The practical implementation is therefore immediate. One splits the trial into $\mathcal{D}_{n,1}$ and $\mathcal{D}_{n,0}$, treats one arm as labeled and the other as unlabeled when estimating a given mean potential outcome, fits and calibrates an arm-specific score, averages the calibrated predictions over the pooled covariate sample, and finally subtracts the two arm-specific estimates to obtain the ATE. In this way, the semisupervised mean estimators in the main text become direct estimators of mean potential outcomes and treatment effects in randomized experiments.

F General Moment Equations Under Missing at Random

We briefly review the general AIPW class for moment-equation targets under missing at random, following Robins et al. [81, 82], Robins and Rotnitzky [80], van der Laan and Robins [114]. Let

$$O = (X, R, RY),$$

where $R \in \{0,1\}$ indicates whether Y is observed, and suppose

$$R \perp Y \mid X, \qquad \pi_0(X) := \mathbb{P}(R=1 \mid X) > 0 \text{ a.s.}$$

Let the target $\theta_0 \in \Theta \subseteq \mathbb{R}^d$ be identified by

$$\mathbb{E}_{P_0}[\varphi(Y,\theta_0)] = 0,$$

where $\varphi : \mathcal{Y} \times \Theta \to \mathbb{R}^d$ is a known identifying function. Define

$$A_0 := \partial_\theta \mathbb{E}_{P_0}[\varphi(Y,\theta)]\big|_{\theta=\theta_0},$$

and assume A_0 is nonsingular.

Theorem 3 (AIPW class for general moment equations under MAR)**.** *For any square-integrable function $m : \mathcal{X} \times \Theta \to \mathbb{R}^d$, define*

$$\psi(O; \theta, m, \pi_0) := m(X, \theta) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}. \tag{24}$$

Then

$$\mathbb{E}[\psi(O; \theta_0, m, \pi_0)] = 0.$$

Hence the equation

$$\mathbb{P}_n \psi(O; \hat{\theta}, m, \pi_0) = 0$$

defines an unbiased AIPW estimating equation for θ_0.

Moreover, let $\hat{\theta}$ be a regular $\sqrt{n}$-consistent solution of this equation, and assume that $\theta \mapsto \mathbb{E}[\psi(O; \theta, m, \pi_0)]$ is differentiable at θ_0 with derivative A_0, and that

$$(\mathbb{P}_n - P_0)\big[\psi(O; \hat{\theta}, m, \pi_0) - \psi(O; \theta_0, m, \pi_0)\big] = o_p\big(n^{-1/2} + \|\hat{\theta} - \theta_0\|\big).$$

Then $\hat{\theta}$ has influence function

$$D_m(O) = -A_0^{-1}\psi(O; \theta_0, m, \pi_0) = -A_0^{-1}\left[m(X, \theta_0) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta_0) - m(X, \theta_0)\}\right]. \tag{25}$$

Thus the class of regular Z-estimators arising from (24) is indexed by the choice of augmentation function m.

The efficient choice in the full MAR model is

$$m_0(X, \theta_0) := \mathbb{E}[\varphi(Y, \theta_0) \mid X], \tag{26}$$

which yields the efficient influence function

$$D_{\text{eff}}(O) = -A_0^{-1}\left[m_0(X, \theta_0) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta_0) - m_0(X, \theta_0)\}\right]. \tag{27}$$

Proof of theorem 3. For any $\theta \in \Theta$,

$$\begin{aligned}
\mathbb{E}\{\psi(O; \theta, m, \pi_0)\} &= \mathbb{E}\left[m(X, \theta) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}\right] \\
&= \mathbb{E}\left[m(X, \theta) + \frac{\mathbb{E}(R \mid X, Y)}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}\right] \\
&= \mathbb{E}\left[m(X, \theta) + \frac{\mathbb{E}(R \mid X)}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}\right] \\
&= \mathbb{E}\big[\varphi(Y, \theta)\big],
\end{aligned}$$

where the third line uses $R \perp Y \mid X$. Evaluating at $\theta = \theta_0$ gives

$$\mathbb{E}\{\psi(O; \theta_0, m, \pi_0)\} = \mathbb{E}\big[\varphi(Y, \theta_0)\big] = 0.$$

Let

$$\Psi_m(\theta) := \mathbb{E}\{\psi(O; \theta, m, \pi_0)\}.$$

The previous calculation shows that $\Psi_m(\theta) = \mathbb{E}[\varphi(Y, \theta)]$ for every θ, so

$$\partial_\theta \Psi_m(\theta)\big|_{\theta=\theta_0} = A_0.$$

Because $\hat{\theta}$ solves the empirical estimating equation,

$$\begin{aligned}
0 &= \mathbb{P}_n \psi(O; \hat{\theta}, m, \pi_0) \\
&= (\mathbb{P}_n - P_0)\psi(O; \theta_0, m, \pi_0) + \{\Psi_m(\hat{\theta}) - \Psi_m(\theta_0)\} \\
&\quad + (\mathbb{P}_n - P_0)\big[\psi(O; \hat{\theta}, m, \pi_0) - \psi(O; \theta_0, m, \pi_0)\big].
\end{aligned}$$

By differentiability of Ψ_m at θ_0,

$$\Psi_m(\hat{\theta}) - \Psi_m(\theta_0) = A_0(\hat{\theta} - \theta_0) + o(\|\hat{\theta} - \theta_0\|).$$

Since $\hat{\theta}$ is $\sqrt{n}$-consistent, the previous display and the stated remainder condition imply

$$A_0(\hat{\theta} - \theta_0) = -(\mathbb{P}_n - P_0)\psi(O; \theta_0, m, \pi_0) + o_p(n^{-1/2}).$$

Multiplying by $-A_0^{-1}$ gives

$$\hat{\theta} - \theta_0 = (\mathbb{P}_n - P_0)D_m(O) + o_p(n^{-1/2}),$$

with D_m given in equation (25). This proves the influence-function formula.

Now define

$$m_0(X, \theta_0) := \mathbb{E}\{\varphi(Y, \theta_0) \mid X\}, \qquad \delta_m(X) := m(X, \theta_0) - m_0(X, \theta_0).$$

Then

$$D_m(O) = D_{\text{eff}}(O) - A_0^{-1}\left(1 - \frac{R}{\pi_0(X)}\right)\delta_m(X).$$

Fix any $u \in \mathbb{R}^d$. Because $\delta_m(X)$ is measurable with respect to X,

$$\mathbb{E}\left[\left(1 - \frac{R}{\pi_0(X)}\right)\delta_m(X) \,\middle|\, X\right] = 0.$$

Also, using $R \perp Y \mid X$ and $\mathbb{E}[\varphi(Y, \theta_0) - m_0(X, \theta_0) \mid X] = 0$,

$$\mathbb{E}\left[\frac{R}{\pi_0(X)}\left(1 - \frac{R}{\pi_0(X)}\right)\{\varphi(Y, \theta_0) - m_0(X, \theta_0)\} \,\middle|\, X\right] = 0.$$

Therefore the cross-term vanishes:

$$\mathbb{E}\left[\left\{u^\top D_{\text{eff}}(O)\right\}\left(1 - \frac{R}{\pi_0(X)}\right)\left\{u^\top A_0^{-1}\delta_m(X)\right\}\right] = 0.$$

Moreover,

$$\mathbb{E}\left[\left(1 - \frac{R}{\pi_0(X)}\right)^2 \,\middle|\, X\right] = \frac{1 - \pi_0(X)}{\pi_0(X)}.$$

Expanding the variance of $u^\top D_m(O)$ therefore yields

$$\text{Var}\{u^\top D_m(O)\} = \text{Var}\{u^\top D_{\text{eff}}(O)\} + \mathbb{E}\left[\frac{1 - \pi_0(X)}{\pi_0(X)}\left\{u^\top A_0^{-1}\delta_m(X)\right\}^2\right].$$

The second term is nonnegative and equals zero if and only if $\delta_m(X) = 0$ almost surely. Hence $m_0(X, \theta_0)$ minimizes the asymptotic variance of every linear contrast, so D_{eff} is the efficient influence function within this class. The missing-data efficiency results of Robins et al. [81, 82], Robins and Rotnitzky [80], van der Laan and Robins [114] identify this same influence function as the semiparametric efficient influence function in the full MAR model. □

Corollary 3 (Two-sample/PPI form)**.** *Suppose the labeled and unlabeled samples arise from the semisupervised model with constant labeling probability ρ_0, and let $S = f(X)$ be a chosen score. Replacing X by S gives the reduced AIPW class*

$$\psi(O; \theta, m, \rho_0) = m(S, \theta) + \frac{R}{\rho_0}\{\varphi(Y, \theta) - m(S, \theta)\}.$$

In the two-sample notation used in the main text, the empirical estimating equation based on the known labeled fraction $\rho_n = n/(n + N)$ can be written as

$$0 = \rho_n \mathbb{P}_n^L\{m(S, \theta)\} + (1 - \rho_n)\mathbb{P}_N^U\{m(S, \theta)\} + \mathbb{P}_n^L\{\varphi(Y, \theta) - m(S, \theta)\}.$$

The efficient choice within the reduced score model is

$$m_0(S, \theta_0) = \mathbb{E}[\varphi(Y, \theta_0) \mid S].$$

Proof. Because the semisupervised design has constant labeling probability, R is independent of (X, Y), hence also independent of Y given $S = f(X)$, and

$$\mathbb{P}(R = 1 \mid S) = \rho_0.$$

Applying theorem 3 with covariate S and observation probability $\pi_0(S) \equiv \rho_0$ gives the displayed reduced AIPW class and the efficient choice

$$m_0(S, \theta_0) = \mathbb{E}[\varphi(Y, \theta_0) \mid S].$$

At the sample level, the two-sample implementation replaces the population labeling probability ρ_0 by the known design fraction $\rho_n = n/(n+N)$. To rewrite the empirical equation in the two-sample notation, note that

$$\mathbb{P}_M\{m(S, \theta)\} = \rho_n \mathbb{P}_n^L\{m(S, \theta)\} + (1 - \rho_n)\mathbb{P}_N^U\{m(S, \theta)\},$$

where $\mathbb{P}_M$ denotes the empirical mean over the pooled sample. Also,

$$\mathbb{P}_M\left[\frac{R}{\rho_n}\{\varphi(Y, \theta) - m(S, \theta)\}\right] = \mathbb{P}_n^L\{\varphi(Y, \theta) - m(S, \theta)\},$$

because $R = 1$ on labeled observations and $R = 0$ on unlabeled observations. Adding the two displays yields exactly

$$0 = \rho_n \mathbb{P}_n^L\{m(S, \theta)\} + (1 - \rho_n)\mathbb{P}_N^U\{m(S, \theta)\} + \mathbb{P}_n^L\{\varphi(Y, \theta) - m(S, \theta)\}.$$

□

G Proof of the representation theorem

Proof of theorem 1. Because equation (5) holds for every $h \in \mathcal{F}$, any empirical balancing weight $\widehat{w}(m_n^\star(X)) \in \mathcal{F}$ satisfies

$$\mathbb{P}_n^L\big[\widehat{w}(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = 0.$$

Adding this zero term to the plug-in estimator equation (3) yields equation (6).

For the second claim, since the identity map belongs to $\mathcal{F}$, taking $f(t) = t$ in equation (7) gives

$$\mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))\, m_n^\star(X)\big] = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1 - \rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\} = \widehat{\psi}_{\mathrm{cal}}.$$

Since $\widehat{w}_n^* \in \mathcal{F}$, equation (5) also gives

$$\mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = 0.$$

Therefore,

$$\mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))Y\big] = \mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))\, m_n^\star(X)\big] + \mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = \widehat{\psi}_{\mathrm{cal}},$$

which is equation (8). □

Lemma 1 (Blockwise residual orthogonality)**.** *Let $B_1, \ldots, B_J$ be the pooled adjacent violator blocks of the isotonic fit, and let $m_{n,\mathrm{iso}}^\star(X_i) = c_j$ for $i \in B_j$. Then*

$$\sum_{i \in B_j} (Y_i - c_j) = 0, \qquad j = 1, \ldots, J.$$

Consequently, for every measurable function $h : \mathbb{R} \to \mathbb{R}$ such that $h(m_{n,\mathrm{iso}}^\star(X_i))$ is finite for all i,

$$\sum_{i=1}^{n} h(m_{n,\mathrm{iso}}^\star(X_i))\{Y_i - m_{n,\mathrm{iso}}^\star(X_i)\} = 0.$$

Proof. The fitted values from isotonic regression are constant on pooled adjacent violator blocks. The KKT conditions imply that residuals sum to zero on each block. Since $h(m_{n,\mathrm{iso}}^\star(X_i)) = h(c_j)$ for all $i \in B_j$,

$$\sum_{i=1}^{n} h(m_{n,\mathrm{iso}}^\star(X_i))\{Y_i - m_{n,\mathrm{iso}}^\star(X_i)\} = \sum_{j=1}^{J} h(c_j) \sum_{i \in B_j} (Y_i - c_j) = 0.$$

□

Proof of proposition 1. The orthogonality claim equation (9) is exactly the conclusion of Lemma 1. For the risk comparison, note that the identity map belongs to $\mathcal{F}_{\text{iso}}$. Since $\hat{f}$ minimizes the empirical squared loss over $\mathcal{F}_{\text{iso}}$,

$$\mathbb{P}_n^L\big[(Y - m^\star_{n,\text{iso}}(X))^2\big] \le \mathbb{P}_n^L\big[(Y - \hat{f}_{\text{id}}(m(X)))^2\big] = \mathbb{P}_n^L\big[(Y - m(X))^2\big],$$

where $\hat{f}_{\text{id}}(t) = t$. □

Proof of proposition 2. The normal equations for equation (10) are

$$\mathbb{P}_n^L\{Y - m^\star_{n,\text{lin}}(X)\} = 0 \qquad \text{and} \qquad \mathbb{P}_n^L\big[m(X)\{Y - m^\star_{n,\text{lin}}(X)\}\big] = 0.$$

Because $m^\star_{n,\text{lin}}(X) = \hat{a}_{\text{lin}} m(X) + \hat{b}_{\text{lin}}$, these imply

$$\mathbb{P}_n^L\big[m^\star_{n,\text{lin}}(X)\{Y - m^\star_{n,\text{lin}}(X)\}\big] = 0.$$

Hence, for any $\hat{w}(X) = a + b\, m^\star_{n,\text{lin}}(X)$,

$$\mathbb{P}_n^L\big[\hat{w}(X)\{Y - m^\star_{n,\text{lin}}(X)\}\big] = a\,\mathbb{P}_n^L\{Y - m^\star_{n,\text{lin}}(X)\} + b\,\mathbb{P}_n^L\big[m^\star_{n,\text{lin}}(X)\{Y - m^\star_{n,\text{lin}}(X)\}\big] = 0.$$

Adding this zero term to equation (11) yields equation (12). □

H Proofs of first-order equivalence between PPI++ and linear calibration

Recall that

$$(\hat{a}_{\text{lin}}, \hat{b}_{\text{lin}}) \in \arg\min_{a,b\in\mathbb{R}} \sum_{i=1}^n \{Y_i - a\, m(X_i) - b\}^2,$$

and define the corresponding linear-calibration estimator by

$$\widehat{\psi}_{\text{lin}} = \rho_n \mathbb{P}_n^L\{\hat{a}_{\text{lin}} m(X) + \hat{b}_{\text{lin}}\} + (1-\rho_n)\mathbb{P}_N^U\{\hat{a}_{\text{lin}} m(\widetilde{X}) + \hat{b}_{\text{lin}}\}.$$

Also let $\widehat{\psi}_{++}$ denote the PPI++ estimator obtained by minimizing the empirical influence-function variance over the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, and let $\hat{\lambda}_{++}$ be the resulting selected coefficient. Define

$$\Delta_n := \mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X),$$

and

$$a_0 := \frac{\text{Cov}\{Y, m(X)\}}{\text{Var}\{m(X)\}}.$$

Proof of Proposition 3. The normal equations for the linear regression fit are

$$\mathbb{P}_n^L\{Y - \hat{a}_{\text{lin}} m(X) - \hat{b}_{\text{lin}}\} = 0$$

and

$$\mathbb{P}_n^L\big[m(X)\{Y - \hat{a}_{\text{lin}} m(X) - \hat{b}_{\text{lin}}\}\big] = 0.$$

The first equation gives

$$\hat{b}_{\text{lin}} = \mathbb{P}_n^L Y - \hat{a}_{\text{lin}} \mathbb{P}_n^L m(X).$$

Substituting this into the second equation yields

$$\hat{a}_{\text{lin}} = \frac{\mathbb{P}_n^L\big[(m(X) - \mathbb{P}_n^L m(X))(Y - \mathbb{P}_n^L Y)\big]}{\mathbb{P}_n^L\big[(m(X) - \mathbb{P}_n^L m(X))^2\big]}.$$

Hence $\hat{a}_{\text{lin}}$ is the ordinary least-squares slope, and under finite second moments,

$$\hat{a}_{\text{lin}} = a_0 + O_p(n^{-1/2}).$$

Using

$$\hat{b}_{\text{lin}} = \mathbb{P}_n^L Y - \hat{a}_{\text{lin}} \mathbb{P}_n^L m(X),$$

we obtain

$$\begin{aligned}
\widehat{\psi}_{\text{lin}} &= \rho_n \mathbb{P}_n^L\{\hat{a}_{\text{lin}} m(X) + \hat{b}_{\text{lin}}\} + (1-\rho_n)\mathbb{P}_N^U\{\hat{a}_{\text{lin}} m(\widetilde{X}) + \hat{b}_{\text{lin}}\} \\
&= \hat{b}_{\text{lin}} + \hat{a}_{\text{lin}}\Big[\rho_n \mathbb{P}_n^L m(X) + (1-\rho_n)\mathbb{P}_N^U m(\widetilde{X})\Big] \\
&= \mathbb{P}_n^L Y - \hat{a}_{\text{lin}}\mathbb{P}_n^L m(X) + \hat{a}_{\text{lin}}\Big[\rho_n \mathbb{P}_n^L m(X) + (1-\rho_n)\mathbb{P}_N^U m(\widetilde{X})\Big] \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{a}_{\text{lin}}\big\{\mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X)\big\} \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{a}_{\text{lin}}\Delta_n.
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\widehat{\psi}_{++}(\hat{\lambda}_{++}) &= \rho_n \mathbb{P}_n^L\{\hat{\lambda}_{++} m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{\hat{\lambda}_{++} m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - \hat{\lambda}_{++} m(X)\} \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{\lambda}_{++}\big\{\mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X)\big\} \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{\lambda}_{++}\Delta_n.
\end{aligned}$$

Therefore,

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = (1-\rho_n)\big(\hat{\lambda}_{++} - \hat{a}_{\text{lin}}\big)\Delta_n.$$

Under the empirical PPI++ variance criterion over the class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, the selected coefficient satisfies

$$\hat{\lambda}_{++} = \frac{\mathbb{P}_n^L\big[(Y - \mathbb{P}_n^L Y)(m(X) - \mathbb{P}_n^L m(X))\big]}{(1-\rho_n)\mathbb{P}_n^L\big[(m(X) - \mathbb{P}_n^L m(X))^2\big] + \rho_n \mathbb{P}_N^U\big[(m(\widetilde{X}) - \mathbb{P}_N^U m(\widetilde{X}))^2\big]}.$$

Its population limit is

$$\lambda_0 = \frac{\text{Cov}\{Y, m(X)\}}{\text{Var}\{m(X)\}} = a_0,$$

so standard M-estimation arguments yield

$$\hat{\lambda}_{++} = a_0 + O_p(n^{-1/2}).$$

Combining this with $\hat{a}_{\text{lin}} = a_0 + O_p(n^{-1/2})$ gives

$$\hat{\lambda}_{++} - \hat{a}_{\text{lin}} = O_p(n^{-1/2}).$$

Next, since $\mathbb{P}_n^L m(X)$ and $\mathbb{P}_N^U m(\widetilde{X})$ are sample means under the same marginal law of X,

$$\Delta_n = O_p(n^{-1/2} + N^{-1/2}).$$

Under the standing assumption $\rho_n = n/(n+N) \to \rho_0 \in (0,1)$, we may write

$$n^{-1/2} = \rho_n^{-1/2}(n+N)^{-1/2}, \qquad N^{-1/2} = (1-\rho_n)^{-1/2}(n+N)^{-1/2},$$

and hence

$$\Delta_n = O_p\Big(\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\}(n+N)^{-1/2}\Big).$$

Therefore,

$$\begin{aligned}
\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} &= (1-\rho_n)\big(\hat{\lambda}_{++} - \hat{a}_{\text{lin}}\big)\Delta_n \\
&= (1-\rho_n)\, O_p(n^{-1/2})\, O_p\Big(\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\}(n+N)^{-1/2}\Big) \\
&= O_p\Big((1-\rho_n)\rho_n^{-1/2}\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\}(n+N)^{-1}\Big).
\end{aligned}$$

Since

$$(1-\rho_n)\rho_n^{-1/2}\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\} = \frac{1-\rho_n}{\rho_n} + \sqrt{\frac{1-\rho_n}{\rho_n}},$$

It follows that

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = O_p\Big(\Big\{\frac{1-\rho_n}{\rho_n} + \sqrt{\frac{1-\rho_n}{\rho_n}}\Big\}(n+N)^{-1}\Big).$$

In particular, this implies the simpler bound

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = O_p\left(\frac{1-\rho_n}{\rho_n}\,(n+N)^{-1}\right).$$

Therefore, under $\rho_n \to \rho_0 \in (0,1)$,

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = O_p\big((n+N)^{-1}\big) = o_p\big((n+N)^{-1/2}\big).$$

This proves the claim. □

Proposition 7 (PPI and AIPW as intercept-only calibration)**.** *Let*

$$\hat{a}_{\text{const}} \in \arg\min_{a\in\mathbb{R}} \sum_{i=1}^{n} \{Y_i - a - m(X_i)\}^2, \qquad m^{\star}_{n,\text{const}}(X) := \hat{a}_{\text{const}} + m(X).$$

Then

$$\hat{a}_{\text{const}} = \mathbb{P}_n^L\{Y - m(X)\}.$$

Consequently, the PPI estimator

$$\widehat{\psi}_{\text{PPI}} := \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}$$

satisfies

$$\widehat{\psi}_{\text{PPI}} = \mathbb{P}_N^U\{m^{\star}_{n,\text{const}}(\widetilde{X})\},$$

while the standard AIPW estimator

$$\widehat{\psi}_{\text{AIPW}} := \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}$$

satisfies

$$\widehat{\psi}_{\text{AIPW}} = \rho_n \mathbb{P}_n^L\{m^{\star}_{n,\text{const}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^{\star}_{n,\text{const}}(\widetilde{X})\}.$$

Hence PPI is the unlabeled-only plug-in estimator based on the mean-calibrated score $m^{\star}_{n,\text{const}}$, whereas AIPW is the corresponding pooled plug-in estimator.

Proof. The objective in the definition of $\hat{a}_{\text{const}}$ is quadratic in a, and its first-order condition is

$$0 = \mathbb{P}_n^L\{Y - a - m(X)\}.$$

Therefore

$$\hat{a}_{\text{const}} = \mathbb{P}_n^L\{Y - m(X)\}.$$

Using $m^{\star}_{n,\text{const}}(\widetilde{X}) = m(\widetilde{X}) + \hat{a}_{\text{const}}$, we obtain

$$\mathbb{P}_N^U\{m^{\star}_{n,\text{const}}(\widetilde{X})\} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \hat{a}_{\text{const}} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\} = \widehat{\psi}_{\text{PPI}}.$$

Likewise,

$$\begin{aligned}
\rho_n \mathbb{P}_n^L\{m^{\star}_{n,\text{const}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^{\star}_{n,\text{const}}(\widetilde{X})\} &= \rho_n \mathbb{P}_n^L\{m(X) + \hat{a}_{\text{const}}\} \\
&\quad + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X}) + \hat{a}_{\text{const}}\} \\
&= \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} \\
&\quad + \hat{a}_{\text{const}} \\
&= \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} \\
&\quad + \mathbb{P}_n^L\{Y - m(X)\} \\
&= \widehat{\psi}_{\text{AIPW}}.
\end{aligned}$$

This proves the claim. □

I Proof of Proposition 5

Proof of proposition 5. Let

$$c := \psi_0 - \mathbb{E}_{P_{0,X}}\{f(X)\}.$$

Because c is constant,

$$\widehat{\psi}(f + c) = \widehat{\psi}(f) + \rho_n c + (1 - \rho_n)c - c = \widehat{\psi}(f).$$

Therefore $\widehat{\psi}(f) = \widehat{\psi}(\tilde{f})$, where

$$\tilde{f}(X) = f(X) + c = f(X) - \mathbb{E}_{P_{0,X}}\{f(X)\} + \psi_0.$$

By construction,

$$\mathbb{E}_{P_{0,X}}\{\tilde{f}(X)\} = \psi_0.$$

Using $\mathbb{E}_{P_0}[Y] = \psi_0$ and $\mathbb{E}_{P_0}\{\tilde{f}(X)\} = \mathbb{E}_{P_{0,X}}\{\tilde{f}(X)\} = \psi_0$, we obtain

$$\begin{aligned}
\widehat{\psi}(f) - \psi_0 &= \widehat{\psi}(\tilde{f}) - \psi_0 \\
&= \rho_n(\mathbb{P}_n^L - P_0)\{\tilde{f}(X)\} + (1 - \rho_n)(\mathbb{P}_N^U - P_{0,X})\{\tilde{f}(\widetilde{X})\} + (\mathbb{P}_n^L - P_0)\{Y - \tilde{f}(X)\} \\
&= \rho_n(\mathbb{P}_n^L - P_0)\Big\{\tilde{f}(X) - \psi_0 + \rho_n^{-1}\big(Y - \tilde{f}(X)\big)\Big\} \\
&\quad + (1 - \rho_n)(\mathbb{P}_N^U - P_{0,X})\big\{\tilde{f}(\widetilde{X}) - \psi_0\big\}.
\end{aligned}$$

Define

$$D_{f,n}^L(X, Y) := \tilde{f}(X) - \psi_0 + \rho_n^{-1}\{Y - \tilde{f}(X)\}, \qquad D_f^U(\widetilde{X}) := \tilde{f}(\widetilde{X}) - \psi_0.$$

Then

$$\widehat{\psi}(f) - \psi_0 = \frac{1}{M}\sum_{i=1}^{n} D_{f,n}^L(X_i, Y_i) + \frac{1}{M}\sum_{j=1}^{N} D_f^U(\widetilde{X}_j).$$

Next,

$$D_{f,n}^L(X, Y) - D_f^L(X, Y) = (\rho_n^{-1} - \rho_0^{-1})\{Y - \tilde{f}(X)\}.$$

Since $\mathbb{E}_{P_0}\{Y - \tilde{f}(X)\} = 0$, it follows that

$$\begin{aligned}
\frac{1}{M}\sum_{i=1}^{n}\{D_{f,n}^L(X_i, Y_i) - D_f^L(X_i, Y_i)\} &= \rho_n(\mathbb{P}_n^L - P_0)\Big[(\rho_n^{-1} - \rho_0^{-1})\{Y - \tilde{f}(X)\}\Big] \\
&= \left(1 - \frac{\rho_n}{\rho_0}\right)(\mathbb{P}_n^L - P_0)\{Y - \tilde{f}(X)\}.
\end{aligned}$$

Because Y and $f(X)$ have finite second moments, $\tilde{f}(X)$ also has finite second moment, so

$$(\mathbb{P}_n^L - P_0)\{Y - \tilde{f}(X)\} = O_p(n^{-1/2}) = O_p(M^{-1/2}).$$

Since $\rho_n \to \rho_0 \in (0, 1)$ by Assumption 1,

$$\frac{1}{M}\sum_{i=1}^{n}\{D_{f,n}^L(X_i, Y_i) - D_f^L(X_i, Y_i)\} = o_p(M^{-1/2}).$$

Therefore

$$\widehat{\psi}(f) - \psi_0 = \frac{1}{M}\sum_{i=1}^{n} D_f^L(X_i, Y_i) + \frac{1}{M}\sum_{j=1}^{N} D_f^U(\widetilde{X}_j) + o_p(M^{-1/2}),$$

which proves the stated asymptotic linearity. The mean-zero property of D_f^L and D_f^U follows from

$$\mathbb{E}_{P_0}[Y] = \mathbb{E}_{P_0}\{\tilde{f}(X)\} = \psi_0.$$

For the asymptotic variance, write

$$\delta(X) := \tilde{f}(X) - \mu_0(X), \qquad \varepsilon := Y - \mu_0(X).$$

Then

$$\mathbb{E}_{P_0}\{\delta(X)\} = 0, \qquad \mathbb{E}_{P_0}(\varepsilon \mid X) = 0, \qquad \mathbb{E}_{P_0}(\varepsilon^2) = \mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big].$$

Also,

$$D_f^U(\widetilde{X}) = \mu_0(\widetilde{X}) - \psi_0 + \delta(\widetilde{X}),$$

and, since $Y - \tilde{f}(X) = \varepsilon - \delta(X)$,

$$\begin{aligned} D_f^L(X,Y) &= \tilde{f}(X) - \psi_0 + \rho_0^{-1}\{Y - \tilde{f}(X)\} \\ &= \mu_0(X) - \psi_0 + \delta(X) + \rho_0^{-1}\{\varepsilon - \delta(X)\} \\ &= \mu_0(X) - \psi_0 + \rho_0^{-1}\varepsilon + \big(1 - \rho_0^{-1}\big)\delta(X). \end{aligned}$$

Set

$$A(X) := \mu_0(X) - \psi_0, \qquad B(X) := \delta(X), \qquad C(X,Y) := \rho_0^{-1}\varepsilon.$$

Then $\mathbb{E}_{P_0}\{C(X,Y) \mid X\} = 0$, so $C(X,Y)$ is uncorrelated with both $A(X)$ and $B(X)$. Hence

$$\mathrm{Var}\{D_f^L(X,Y)\} = \mathrm{Var}\big\{A(X) + \big(1 - \rho_0^{-1}\big)B(X)\big\} + \mathrm{Var}\{C(X,Y)\},$$

while

$$\mathrm{Var}\{D_f^U(\widetilde{X})\} = \mathrm{Var}\{A(\widetilde{X}) + B(\widetilde{X})\}.$$

Therefore

$$\begin{aligned} &\rho_0\mathrm{Var}\{D_f^L(X,Y)\} + (1-\rho_0)\mathrm{Var}\{D_f^U(\widetilde{X})\} \\ &= \rho_0\mathrm{Var}\big\{A + \big(1 - \rho_0^{-1}\big)B\big\} + (1-\rho_0)\mathrm{Var}(A+B) + \rho_0\mathrm{Var}(C). \end{aligned}$$

Expanding the two variances involving A and B gives

$$\begin{aligned} &\rho_0\mathrm{Var}\big\{A + \big(1 - \rho_0^{-1}\big)B\big\} + (1-\rho_0)\mathrm{Var}(A+B) \\ &= \{\rho_0 + (1-\rho_0)\}\mathrm{Var}(A) \\ &\quad + \Big\{\rho_0\big(1 - \rho_0^{-1}\big)^2 + (1-\rho_0)\Big\}\,\mathrm{Var}(B) \\ &\quad + 2\,\Big\{\rho_0\big(1 - \rho_0^{-1}\big) + (1-\rho_0)\Big\}\,\mathrm{Cov}(A,B). \end{aligned}$$

Now

$$1 - \rho_0^{-1} = -\frac{1-\rho_0}{\rho_0},$$

so

$$\rho_0\big(1 - \rho_0^{-1}\big)^2 + (1-\rho_0) = \frac{(1-\rho_0)^2}{\rho_0} + (1-\rho_0) = \frac{1-\rho_0}{\rho_0},$$

and

$$\rho_0\big(1 - \rho_0^{-1}\big) + (1-\rho_0) = -(1-\rho_0) + (1-\rho_0) = 0.$$

Thus

$$\rho_0\mathrm{Var}\big\{A + \big(1 - \rho_0^{-1}\big)B\big\} + (1-\rho_0)\mathrm{Var}(A+B) = \mathrm{Var}(A) + \frac{1-\rho_0}{\rho_0}\mathrm{Var}(B).$$

Also,

$$\rho_0\mathrm{Var}(C) = \rho_0 \cdot \rho_0^{-2}\mathbb{E}_{P_0}(\varepsilon^2) = \rho_0^{-1}\mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big].$$

Since $\mathbb{E}_{P_0}\{B(X)\} = 0$,

$$\mathrm{Var}(B) = \mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big].$$

Combining the preceding displays yields

$$\begin{aligned} \rho_0\mathrm{Var}\{D_f^L(X,Y)\} + (1-\rho_0)\mathrm{Var}\{D_f^U(\widetilde{X})\} &= \mathrm{Var}\{\mu_0(X)\} \\ &\quad + \rho_0^{-1}\mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big] \\ &\quad + \frac{1-\rho_0}{\rho_0}\mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big]. \end{aligned}$$

This expression is minimized when $\tilde{f}(X) = \mu_0(X)$ almost surely, in particular at $f = \mu_0$. Standard semiparametric characterization results for missing-data models imply that this minimum is the semiparametric efficiency bound in the unrestricted two-sample model; see, for example, Bickel et al. [10], Robins et al. [82], Robins and Rotnitzky [80], van der Laan and Robins [114], Tsiatis [101]. This completes the proof. □

Lemma 2 (Bounded variation after isotonic post-processing)**.** *Suppose Assumption 3 holds. Let $T := m(X)$ and $S_n := m^\star_{n,\mathrm{iso}}(X)$. Define*

$$\eta(t) := \mathbb{E}_{P_0}[Y \mid T = t], \qquad \eta_n(s) := \mathbb{E}_{P_0}[Y \mid S_n = s].$$

If $t \mapsto \eta(t)$ has total variation at most V, then, conditional on the labeled sample, the map $s \mapsto \eta_n(s)$ has total variation at most V. Moreover, $|\eta_n(s)| \le C_0$ on the support of S_n.

Proof. We adapt the proof of Lemma 6 in [106]. Let $S_n = f_n(T)$, where f_n is the fitted isotonic map. Conditional on the labeled sample, f_n is nondecreasing and piecewise constant. Hence, for each s in the support of S_n, there exists an interval $B_s \subset \mathbb{R}$ such that

$$\{S_n = s\} = \{T \in B_s\}.$$

Therefore

$$\eta_n(s) = \mathbb{E}_{P_0}[Y \mid S_n = s] = \mathbb{E}_{P_0}[\eta(T) \mid T \in B_s].$$

Let $\eta = \eta^+ - \eta^-$ be the Jordan decomposition of η, where η^+ and η^- are nondecreasing and

$$\|\eta\|_{TV} = \|\eta^+\|_{TV} + \|\eta^-\|_{TV} \le V.$$

Then

$$\eta_n(s) = \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s] - \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s].$$

If $s_1 < s_2$, then monotonicity of f_n implies every point of B_{s_1} is no larger than every point of B_{s_2}. Hence the conditional distribution of $T \mid T \in B_{s_2}$ first-order stochastically dominates that of $T \mid T \in B_{s_1}$. Therefore, for any nondecreasing function g,

$$\mathbb{E}_{P_0}[g(T) \mid T \in B_{s_1}] \le \mathbb{E}_{P_0}[g(T) \mid T \in B_{s_2}].$$

Applying this with $g = \eta^+$ and $g = \eta^-$, both maps

$$s \mapsto \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s] \qquad \text{and} \qquad s \mapsto \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s]$$

are nondecreasing.

Moreover, each of these conditional expectations takes values in the range of the corresponding function, so

$$\big\|s \mapsto \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s]\big\|_{TV} \le \|\eta^+\|_{TV}, \qquad \big\|s \mapsto \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s]\big\|_{TV} \le \|\eta^-\|_{TV}.$$

Hence

$$\|\eta_n\|_{TV} \le \|\eta^+\|_{TV} + \|\eta^-\|_{TV} = \|\eta\|_{TV} \le V.$$

Finally, if $|\eta(t)| \le C_0$ on the support of T, then

$$|\eta_n(s)| = |\mathbb{E}_{P_0}[\eta(T) \mid T \in B_s]| \le C_0$$

for all s in the support of S_n. □

Lemma 3 (Empirical calibration centers the limit)**.** *Suppose Assumptions 1 and 2 hold and the calibration score equations contain the constant score $h \equiv 1$. Then*

$$\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0.$$

Proof. Let $T := m(X)$, and write $m_0 = f_0(T)$, where

$$f_0 \in \arg\min_{f \in \mathcal{F}_{\mathrm{iso}}} \mathbb{E}_{P_0}\big[(Y - f(T))^2\big].$$

Because $\mathcal{F}_{\mathrm{iso}}$ is closed under addition of constants, for every $c \in \mathbb{R}$ the function $f_0 + c$ also belongs to $\mathcal{F}_{\mathrm{iso}}$. Hence the function

$$g(c) := \mathbb{E}_{P_0}\big[(Y - f_0(T) - c)^2\big]$$

is minimized at $c = 0$. Since g is differentiable,

$$0 = g'(0) = -2\,\mathbb{E}_{P_0}[Y - f_0(T)].$$

Therefore

$$\mathbb{E}_{P_0}[Y] = \mathbb{E}_{P_0}[f_0(T)].$$

Recalling that $m_0(X) = f_0(m(X)) = f_0(T)$ and $\psi_0 = \mathbb{E}_{P_0}[Y]$, we conclude

$$\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0.$$

□

Proof convention. In the following proofs, fix $\delta > 0$. By $\sup_x |m^\star_{n,\mathrm{iso}}(x)| = O_p(1)$, there exist a deterministic constant $C = C_\delta \geq C_0$ and $n_0 < \infty$ such that the event

$$A_{n,C} := \left\{ \sup_x |m^\star_{n,\mathrm{iso}}(x)| \leq C \right\}$$

satisfies $P_0(A_{n,C}) \geq 1 - \delta$ for all $n \geq n_0$. We carry out the deterministic bounded-class arguments on $A_{n,C}$, where all implicit constants may depend on C but not on n. Since δ is arbitrary, the resulting rates and distributional statements hold unconditionally.

Lemma 4 (Isotonic L^2 rate)**.** *Suppose Assumptions 1 and 2 holds. Then*

$$\|m^\star_{n,\mathrm{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

Proof of Lemma 4. Let $T := m(X)$ and let

$$\mathcal{M}_C := \{f(T) : f \in \mathcal{F}_{\mathrm{iso}},\ \|f(T)\|_\infty \leq C\}.$$

By Assumption 2, $g_0(T)$ is bounded by $C_0 \leq C$, and on $A_{n,C}$ we have $m^\star_{n,\mathrm{iso}} \in \mathcal{M}_C$. Since clipping any monotone candidate to $[-C, C]$ preserves monotonicity and can only decrease squared risk against $g_0(T)$, the population isotonic projection m_0 also belongs to $\mathcal{M}_C$. Thus the problem is a one-dimensional isotonic least-squares regression of Y on the score T. Let

$$g_0(T) := \mathbb{E}_{P_0}[Y \mid T].$$

Then, for every $m \in \mathcal{M}_C$,

$$P_0\{(Y - m)^2\} = P_0\{(Y - g_0)^2\} + P_0\{(g_0 - m)^2\},$$

which shows that m_0 is the $L^2(P_T)$-projection of g_0 onto the closed convex set $\mathcal{M}_C$.

Now define the centered class

$$\mathcal{F}_0 := \{g = m - m_0 :\ m \in \mathcal{M}_C\}.$$

For $g \in \mathcal{F}_0$, let $\ell_g(Z) := (Y - m_0(T) - g(T))^2 - (Y - m_0(T))^2$, where $Z = (X, Y)$. Since $m^\star_{n,\mathrm{iso}}$ minimizes the empirical squared risk over $\mathcal{F}_{\mathrm{iso}}$, on $A_{n,C}$ the random element $\hat{g} := m^\star_{n,\mathrm{iso}} - m_0$ belongs to $\mathcal{F}_0$ and satisfies

$$\mathbb{P}^L_n \ell_{\hat{g}} \leq 0 = \mathbb{P}^L_n \ell_0.$$

Moreover,

$$P_0 \ell_g = \|g\|^2_{2,P_{0,X}} - 2P_0\left[(Y - m_0)g\right].$$

Because g is measurable with respect to T, the tower property gives

$$P_0\left[(Y - m_0)g\right] = P_0\left[(g_0 - m_0)g\right].$$

Since m_0 is the $L^2(P_T)$-projection of g_0 onto the convex set $\mathcal{M}_C$, the Hilbert-space projection inequality yields

$$P_0\left[(g_0 - m_0)(m - m_0)\right] \leq 0 \qquad \text{for every } m \in \mathcal{M}_C.$$

Hence

$$P_0 \ell_g \geq \|g\|^2_{2,P_{0,X}},$$

so squared loss has the required quadratic margin around m_0.

We next bound the modulus of continuity of the empirical process indexed by the localized classes

$$\mathcal{F}_\delta := \{g \in \mathcal{F}_0 : \|g\|_{2,P_{0,X}} \leq \delta\}.$$

Because every $m \in \mathcal{M}_C$ is bounded by C, every $g \in \mathcal{F}_\delta$ is uniformly bounded, and the standard bracketing bound for one-dimensional monotone classes gives

$$\log N_{[]}(\varepsilon, \mathcal{F}_\delta, L_2(P_{0,X})) \lesssim \delta/\varepsilon, \qquad 0 < \varepsilon < \delta.$$

Therefore

$$J_{[]}(\delta, \mathcal{F}_\delta, L_2(P_{0,X})) \lesssim \delta^{1/2};$$

see, e.g., van der Vaart and Wellner [118].

Write $\xi := Y - m_0(T)$, so that $\ell_g = g^2 - 2\xi g$. The class $\{g^2 : g \in \mathcal{F}_\delta\}$ is a bounded Lipschitz image of $\mathcal{F}_\delta$, so by Lemma 3.4.2 of van der Vaart and Wellner [118] and the same entropy bound,

$$E^* \sup_{g \in \mathcal{F}_\delta} \sqrt{n}\, |(\mathbb{P}_n^L - P_0)(g^2)| \lesssim \delta^{1/2}.$$

For the multiplier term, Assumption 2 implies that $Y - g_0(T)$ is sub-Gaussian or subexponential. Since both $g_0(T)$ and $m_0(T)$ are bounded by C, the difference $g_0(T) - m_0(T)$ is bounded, and therefore $\xi = Y - m_0(T)$ is also sub-Gaussian or subexponential. Thus the product class $\xi\mathcal{F}_\delta := \{\xi g : g \in \mathcal{F}_\delta\}$ satisfies the same localized bracketing bound in the Bernstein-type norm used in the subexponential least-squares proof of Bibaut and van der Laan [9], yielding

$$E^* \sup_{g \in \mathcal{F}_\delta} \sqrt{n}\, |(\mathbb{P}_n^L - P_0)(\xi g)| \lesssim \delta^{1/2}.$$

Combining the previous two displays,

$$E^* \sup_{g \in \mathcal{F}_\delta} \sqrt{n}\, |(\mathbb{P}_n^L - P_0)\ell_g| \lesssim \delta^{1/2}.$$

We may therefore apply Theorem 3.4.1 of van der Vaart and Wellner [118] with $d(g) = \|g\|_{2,P_{0,X}}$, quadratic margin $P_0\ell_g \gtrsim d^2(g)$, and modulus $\phi_n(\delta) \asymp \delta^{1/2}$. The fixed-point condition $r_n^2 \phi_n(r_n^{-1}) \lesssim \sqrt{n}$ gives $r_n \asymp n^{1/3}$, and hence

$$\|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}} = O_p(n^{-1/3}), \qquad \|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

The preceding display is obtained on $A_{n,C}$. Since $P_0(A_{n,C}^c) \le \delta$ for all large n, and $\delta > 0$ was arbitrary, the same rate holds unconditionally. This is the classical one-dimensional isotonic rate, stated directly in the random-design $L^2(P_{0,X})$ norm relevant for the theorem. □

Theorem 4 (Calibration error of isotonic post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\widehat{m}_0(X) := \mathbb{E}_{P_0}[Y \mid m_{n,\mathrm{iso}}^\star(X)].$$

Then

$$\|\widehat{m}_0 - m_{n,\mathrm{iso}}^\star\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

Proof. The proof follows that of Theorem 1 in [106]; see also [102]. Let

$$T := m(X), \qquad S_n := m_{n,\mathrm{iso}}^\star(X), \qquad \eta_n(s) := \mathbb{E}_{P_0}[Y \mid S_n = s],$$

and define

$$\Delta_n(X) := \widehat{m}_0(X) - m_{n,\mathrm{iso}}^\star(X) = \eta_n(S_n) - S_n.$$

Conditional on the labeled sample, $h_n(s) := \eta_n(s) - s$ is a measurable function of s. Applying proposition 1 pathwise with $h = h_n$ yields

$$\mathbb{P}_n^L[\Delta_n(X)\{Y - S_n\}] = 0.$$

Also,

$$P_0[\Delta_n(X)\{Y - S_n\}] = P_0[\Delta_n(X)\,\mathbb{E}_{P_0}[Y - S_n \mid S_n]] = P_0(\Delta_n^2) = \|\Delta_n\|_{2,P_{0,X}}^2.$$

Hence

$$\|\Delta_n\|_{2,P_{0,X}}^2 = (P_0 - \mathbb{P}_n^L)[\Delta_n(X)\{Y - S_n\}]\,.$$

Fix $\delta > 0$, and work on the event $A_{n,C}$ from the proof convention. On this event, S_n is uniformly bounded by C. By Lemma 2, the function η_n belongs to the bounded-variation class

$$\mathcal{F}_{\mathrm{TV},C} := \{f : \mathbb{R} \to \mathbb{R} : \|f\|_\infty \le C,\ \|f\|_{TV} \le V\},$$

where $V < \infty$ is the variation bound in Assumption 3. Let

$$\mathcal{F}_{\mathrm{iso},C} := \{f : \mathbb{R} \to \mathbb{R} : f \text{ nondecreasing and } \|f\|_\infty \le C\}.$$

Since $S_n = f_n(T)$ for a nondecreasing fitted isotonic map f_n, and composition with a monotone map preserves bounded variation, the function

$$t \mapsto \eta_n\{f_n(t)\}$$

also belongs to $\mathcal{F}_{\mathrm{TV},C}$, after enlarging C if necessary. Thus $\eta_n(S_n) = h_1(T)$ for some $h_1 = f_1 \circ T$ with $f_1 \in \mathcal{F}_{\mathrm{TV},C}$, while $S_n = h_2(T)$ for $h_2 = f_2 \circ T$ with $f_2 = f_n \in \mathcal{F}_{\mathrm{iso},C}$.

Define the localized product class

$$\mathcal{G}_{n,C,\delta} := \Big\{ (h_1 - h_2)(Y - h_2) : \begin{array}{l} h_1 = f_1 \circ T,\ h_2 = f_2 \circ T, \\ f_1 \in \mathcal{F}_{\mathrm{TV},C},\ f_2 \in \mathcal{F}_{\mathrm{iso},C}, \\ \|h_1 - h_2\|_{2,P_{0,X}} \le \delta \end{array} \Big\}.$$

Then

$$\Delta_n(X)\{Y - S_n\} \in \mathcal{G}_{n,C,\delta_n}, \qquad \delta_n := \|\Delta_n\|_{2,P_{0,X}}.$$

Because bounded-variation functions are differences of bounded monotone functions, standard entropy-preservation results for Lipschitz transformations imply that the localized entropy integral of $\mathcal{G}_{n,C,\delta}$ satisfies

$$J_{[]}(\delta, \mathcal{G}_{n,C,\delta}, L_2(P_0)) \lesssim \delta^{1/2}.$$

Moreover, Assumption 2 implies that $Y - \eta(T)$ is sub-Gaussian or subexponential, and on $A_{n,C}$ the difference $\eta(T) - S_n$ is bounded. Therefore $Y - S_n$ has the same tail type, up to constants. The same localized multiplier empirical-process argument used in the proof of Lemma 4 then gives

$$E^* \sup_{g \in \mathcal{G}_{n,C,\delta}} |(\mathbb{P}_n^L - P_0)g| \lesssim n^{-1/2}\delta^{1/2}.$$

Set $\varepsilon_n := n^{-1/3}$. For $s \ge 0$, let

$$B_s := \{2^s \varepsilon_n \le \delta_n < 2^{s+1}\varepsilon_n\}.$$

On $B_s \cap A_{n,C}$,

$$2^{2s}\varepsilon_n^2 \le \delta_n^2 \le \sup_{g \in \mathcal{G}_{n,C,2^{s+1}\varepsilon_n}} |(\mathbb{P}_n^L - P_0)g|.$$

Therefore, by Markov's inequality,

$$P_0(B_s \cap A_{n,C}) \lesssim \frac{E^* \sup_{g \in \mathcal{G}_{n,C,2^{s+1}\varepsilon_n}} |(\mathbb{P}_n^L - P_0)g|}{2^{2s}\varepsilon_n^2} \lesssim \frac{n^{-1/2}(2^{s+1}\varepsilon_n)^{1/2}}{2^{2s}\varepsilon_n^2} \lesssim 2^{-3s/2},$$

since $n^{-1/2}\varepsilon_n^{-3/2} = 1$. Summing over $s \ge S$ yields

$$P_0(\delta_n \ge 2^S \varepsilon_n,\ A_{n,C}) \lesssim \sum_{s=S}^{\infty} 2^{-3s/2} \xrightarrow[S \to \infty]{} 0.$$

Because $P_0(A_{n,C}^c) \le \delta$ for all large n, and $\delta > 0$ is arbitrary, it follows that

$$\|\Delta_n\|_{2,P_{0,X}} = O_p(n^{-1/3}), \qquad \|\Delta_n\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

This is exactly the claimed calibration-error rate. □

Proof of Lemma 3. Because $h \equiv 1$ belongs to the calibration score class, equation (5) gives

$$\mathbb{P}_n^L\{Y - m_{n,\mathrm{iso}}^\star(X)\} = 0.$$

Hence

$$\mathbb{P}_n^L Y - \mathbb{P}_n^L m_0(X) = \mathbb{P}_n^L\{m_{n,\mathrm{iso}}^\star(X) - m_0(X)\}.$$

By Cauchy–Schwarz and Lemma 4,

$$P_{0,X}\left|m_{n,\mathrm{iso}}^\star - m_0\right| \le \|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}} = O_p(n^{-1/3}),$$

and Lemma 5 gives

$$(\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = O_p(n^{-2/3}).$$

Therefore

$$\mathbb{P}_n^L\{m_{n,\mathrm{iso}}^\star(X) - m_0(X)\} = P_{0,X}\{m_{n,\mathrm{iso}}^\star - m_0\} + (\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = o_p(1).$$

By Assumption 1, the law of large numbers yields $\mathbb{P}_n^L Y \to_p \psi_0$ and $\mathbb{P}_n^L m_0(X) \to_p \mathbb{E}_{P_{0,X}}[m_0(X)]$. Therefore $\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0$. □

Lemma 5 (Centered empirical-process bounds for isotonic calibration)**.** *Suppose Assumptions 1 and 2 holds. Then*

$$(\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star(X) - m_0(X)\} = O_p(n^{-2/3}),$$
$$(\mathbb{P}_N^U - P_{0,X})\{m_{n,\mathrm{iso}}^\star(\widetilde{X}) - m_0(\widetilde{X})\} = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Proof of Lemma 5. By Lemma 4,

$$\|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

To control the centered empirical-process term, work on the event $A_{n,C}$ from the proof convention and consider the localized class

$$\mathcal{F}_{n,C} := \{g = f \circ m - m_0 : \ f \in \mathcal{F}_{\mathrm{iso}}, \ \|f \circ m\|_\infty \le C, \ \|g\|_{2,P_{0,X}} \lesssim n^{-1/3}\}.$$

On $A_{n,C}$, the function $m_{n,\mathrm{iso}}^\star - m_0$ belongs to $\mathcal{F}_{n,C}$. Because m is fixed, this is a localized class of bounded monotone functions of a one-dimensional score. Standard bracketing entropy bounds for monotone classes imply

$$J_{[]}(r, \mathcal{F}_{n,C}, L_2(P_0)) \lesssim r^{1/2}.$$

Applying a local maximal inequality for empirical processes, such as Lemma 3.4.2 of van der Vaart and Wellner [118], at radius $r_n \asymp n^{-1/3}$ yields

$$\sup_{g \in \mathcal{F}_{n,C}} |(\mathbb{P}_n^L - P_{0,X})g| = O_p\Big(n^{-1/2} r_n^{1/2}\Big) = O_p(n^{-2/3}).$$

In particular,

$$(\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = O_p(n^{-2/3}).$$

Since $P_0(A_{n,C}^c) \le \delta$ for all large n, and $\delta > 0$ is arbitrary, the same bound holds unconditionally.

For the unlabeled sample, conditional on the labeled data, $m_{n,\mathrm{iso}}^\star - m_0$ is fixed and the unlabeled sample is independent, so

$$(\mathbb{P}_N^U - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = O_p\left(N^{-1/2}\|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big)$$

by Lemma 4. □

Lemma 6 (Centered empirical-process bounds for isotonic post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\Delta_n(X) := \widehat{m}_0(X) - m_{n,\mathrm{iso}}^\star(X).$$

Then

$$(\mathbb{P}_n^L - P_{0,X})\Delta_n(X) = O_p(n^{-2/3}), \qquad (\mathbb{P}_N^U - P_{0,X})\Delta_n(\widetilde{X}) = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Proof. By theorem 4,

$$\|\Delta_n\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

Let $V < \infty$ be the variation bound in Assumption 3. Fix $\delta > 0$, and work on the event $A_{n,C}$. On this event, $m_{n,\mathrm{iso}}^\star$ is uniformly bounded by C. By Lemma 2, the function

$$s \mapsto \eta_n(s) := \mathbb{E}_{P_0}[Y \mid m_{n,\mathrm{iso}}^\star(X) = s]$$

has total variation at most V and is bounded by C on the support of $m_{n,\mathrm{iso}}^\star(X)$. Since $m_{n,\mathrm{iso}}^\star(X) = f_n(T)$ with $T = m(X)$ and f_n nondecreasing, it follows that

$$\widehat{m}_0(X) = \eta_n\{f_n(T)\}$$

is a bounded-variation transform of T, while $m_{n,\mathrm{iso}}^\star(X)$ is a bounded monotone transform of T.

Consider the classes

$$\mathcal{F}_{\mathrm{TV},C} := \{f : \mathbb{R} \to \mathbb{R} : \|f\|_\infty \le C, \ \|f\|_{TV} \le V\},$$
$$\mathcal{F}_{\mathrm{iso},C} := \{f : \mathbb{R} \to \mathbb{R} : f \text{ nondecreasing and } \|f\|_\infty \le C\},$$

and define

$$\mathcal{H}_{n,C} := \Big\{ h_1 - h_2 : \begin{array}{l} h_1 = f_1 \circ T,\ h_2 = f_2 \circ T, \\ f_1 \in \mathcal{F}_{\mathrm{TV},C},\ f_2 \in \mathcal{F}_{\mathrm{iso},C}, \\ \|h_1 - h_2\|_{2,P_{0,X}}^2 \lesssim n^{-2/3} \end{array} \Big\}.$$

On $A_{n,C}$, the random function Δ_n belongs to $\mathcal{H}_{n,C}$.

Because bounded-variation functions are differences of bounded monotone functions, the localized bracketing entropy of $\mathcal{H}_{n,C}$ obeys the same bound as in the proof of theorem 4, namely

$$J_{[]}(\delta, \mathcal{H}_{n,C}, L_2(P_0)) \lesssim \delta^{1/2}.$$

Applying the same local maximal inequality at localization radius $n^{-1/3}$ therefore gives, on $A_{n,C}$,

$$|(\mathbb{P}_n^L - P_{0,X})\Delta_n(X)| = O_p(n^{-2/3}).$$

Since $P_0(A_{n,C}^c) \le \delta$ for all large n, and $\delta > 0$ was arbitrary, the same bound holds unconditionally.

For the unlabeled sample, conditional on the labeled data, Δ_n is fixed and the unlabeled sample is independent. Hence

$$(\mathbb{P}_N^U - P_{0,X})\Delta_n(\widetilde{X}) = O_p\left(N^{-1/2}\|\Delta_n\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big)$$

by theorem 4. □

J Proof of asymptotic linearity

Proof of theorem 2. The proof combines the exact AIPW representation from Theorem 1 with a localized empirical-process bound for the isotonic calibrator class. The latter controls the remainder generated by estimating the monotone post-processing map and yields the $n^{-2/3}$ remainder rate. The overall argument is closely related to the calibrated DML framework of van der Laan et al. [108]; see also Kennedy [53] for background on semiparametric influence-function arguments.

By Lemma 3, $P_{0,X} m_0 = \psi_0$. Using theorem 1 with $\widehat{w} \equiv 1$,

$$\widehat{\psi}_{\mathrm{iso}} = \rho_n \mathbb{P}_n^L\{m_{n,\mathrm{iso}}^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_{n,\mathrm{iso}}^\star(\widetilde{X})\} + \rho_n \mathbb{P}_n^L\{Y - m_{n,\mathrm{iso}}^\star(X)\}.$$

Add and subtract the AIPW form built from m_0:

$$\widehat{\psi}_{\mathrm{iso}} = \rho_n \mathbb{P}_n^L\left\{m_0 + \rho_n^{-1}(Y - m_0)\right\} + (1-\rho_n)\mathbb{P}_N^U\{m_0(\widetilde{X})\} + (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m_{n,\mathrm{iso}}^\star - m_0\}.$$

Subtracting ψ_0 gives

$$\begin{aligned} \widehat{\psi}_{\mathrm{iso}} - \psi_0 = \rho_n(\mathbb{P}_n^L - P_0)\Big\{m_0 - \psi_0 + \rho_n^{-1}(Y - m_0)\Big\} \\ + (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\{m_0 - \psi_0\} \\ + (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m_{n,\mathrm{iso}}^\star - m_0\}. \end{aligned}$$

To replace ρ_n^{-1} by ρ_0^{-1}, note that

$$\rho_n(\mathbb{P}_n^L - P_0)\Big\{(\rho_n^{-1} - \rho_0^{-1})(Y - m_0)\Big\} = \left(1 - \frac{\rho_n}{\rho_0}\right)(\mathbb{P}_n^L - P_0)(Y - m_0).$$

The right-hand side is $o_p(M^{-1/2})$ because $\rho_n \to \rho_0$ and

$$(\mathbb{P}_n^L - P_0)(Y - m_0) = O_p(n^{-1/2}) = O_p(M^{-1/2}).$$

Also,

$$\begin{aligned} (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m_{n,\mathrm{iso}}^\star - m_0\} = (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} \\ - (1-\rho_n)(\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\}. \end{aligned}$$

By Lemma 5, the labeled term is $O_p(n^{-2/3})$ and the unlabeled term is $O_p(n^{-1/3}N^{-1/2})$. Thus the remainder $R_{n,N}$, defined as the difference between $\widehat{\psi}_{\mathrm{iso}} - \psi_0$ and the leading empirical-process terms in equation (14), satisfies the bound stated in theorem 2. Since $n^{-2/3} = o(M^{-1/2})$ and $n^{-1/3}N^{-1/2} = o(M^{-1/2})$, this yields equation (14). □

Proof of corollary 1. By theorem 2,

$$\sqrt{M}\,(\widehat{\psi}_{\text{iso}} - \psi_0) = \frac{1}{\sqrt{M}} \sum_{i=1}^{n} D^L_{m_0}(X_i, Y_i) + \frac{1}{\sqrt{M}} \sum_{j=1}^{N} D^U_{m_0}(\widetilde{X}_j) + o_p(1).$$

The labeled and unlabeled samples are independent, each summand has mean zero, and $\rho_n \to \rho_0 \in (0,1)$ by Assumption 1. A standard triangular-array central limit theorem therefore yields the stated asymptotic normal law with variance σ_0^2.

For the variance estimator, let $g_n := m^\star_{n,\text{iso}} - m_0$. By Lemma 4,

$$P_{0,X}(g_n^2) = \|g_n\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

Moreover, the proof of Lemma 4 already established the localized empirical-process bound

$$\sup_{g \in \mathcal{F}_\delta} |(\mathbb{P}^L_n - P_{0,X})(g^2)| = O_p(n^{-1/2}\delta^{1/2}),$$

for the squared class $\{g^2 : g \in \mathcal{F}_\delta\}$. Taking $\delta \asymp n^{-1/3}$, which matches the localization radius of g_n, gives

$$(\mathbb{P}^L_n - P_{0,X})(g_n^2) = O_p(n^{-2/3}).$$

Hence

$$\mathbb{P}^L_n(g_n^2) = O_p(n^{-2/3}) = o_p(1).$$

On the boundedness event $A_{n,C}$, conditional on the labeled data, g_n is fixed and $|g_n| \le 2C$, so

$$(\mathbb{P}^U_N - P_{0,X})(g_n^2) = O_p\left(N^{-1/2}\|g_n^2\|_{2,P_{0,X}}\right) = O_p\left(C\,N^{-1/2}\|g_n\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Therefore

$$\mathbb{P}^U_N(g_n^2) = O_p(n^{-2/3}) + O_p\big(n^{-1/3}N^{-1/2}\big) = o_p(1).$$

Since $P_0(A^c_{n,C}) \le \delta$ for all large n and arbitrary $\delta > 0$, these bounds hold unconditionally. Also, $\widehat{\psi}_{\text{iso}} \to_p \psi_0$ follows from theorem 2.

Finally,

$$\widehat{D}^L - D^L_{m_0} = (1 - \rho_n^{-1})g_n - (\widehat{\psi}_{\text{iso}} - \psi_0) + (\rho_n^{-1} - \rho_0^{-1})(Y - m_0),$$

and

$$\widehat{D}^U - D^U_{m_0} = g_n(\widetilde{X}) - (\widehat{\psi}_{\text{iso}} - \psi_0).$$

Because $\rho_n \to \rho_0$, $Y - m_0$ has finite second moment, and $\mathbb{P}^L_n(g_n^2), \mathbb{P}^U_N(g_n^2) = o_p(1)$, it follows that

$$\mathbb{P}^L_n\Big[\{\widehat{D}^L - D^L_{m_0}\}^2\Big] = o_p(1), \qquad \mathbb{P}^U_N\Big[\{\widehat{D}^U - D^U_{m_0}\}^2\Big] = o_p(1).$$

Therefore $\widehat{\sigma}^2 \to_p \sigma_0^2$. The Wald interval claim then follows by Slutsky's theorem. □

K Proof of post-processing equivalence and reduced-model efficiency

Proof of proposition 4. Let

$$\Delta_n(X) := \widehat{m}_0(X) - m^\star_{n,\text{iso}}(X).$$

Using the definition of the AIPW class together with $\mathbb{P}^L_n\{Y - m^\star_{n,\text{iso}}(X)\} = 0$, we have

$$\begin{aligned}
\widehat{\psi}_{\text{eff}} - \widehat{\psi}_{\text{iso}} &= \rho_n \mathbb{P}^L_n \Delta_n(X) + (1-\rho_n)\mathbb{P}^U_N \Delta_n(\widetilde{X}) + \mathbb{P}^L_n\{Y - \widehat{m}_0(X)\} - \mathbb{P}^L_n\{Y - m^\star_{n,\text{iso}}(X)\} \\
&= \rho_n \mathbb{P}^L_n \Delta_n(X) + (1-\rho_n)\mathbb{P}^U_N \Delta_n(\widetilde{X}) - \mathbb{P}^L_n \Delta_n(X) \\
&= (1-\rho_n)(\mathbb{P}^U_N - \mathbb{P}^L_n)\Delta_n.
\end{aligned}$$

Now decompose

$$(1-\rho_n)(\mathbb{P}^U_N - \mathbb{P}^L_n)\Delta_n = (1-\rho_n)(\mathbb{P}^U_N - P_{0,X})\Delta_n - (1-\rho_n)(\mathbb{P}^L_n - P_{0,X})\Delta_n.$$

By Lemma 6, the labeled term is $O_p(n^{-2/3})$ and the unlabeled term is $O_p(n^{-1/3}N^{-1/2})$. Since $\rho_n \to \rho_0 \in (0,1)$ by Assumption 1, both rates are $o_p(M^{-1/2})$. Therefore

$$\widehat{\psi}_{\text{eff}} - \widehat{\psi}_{\text{iso}} = (1-\rho_n)(\mathbb{P}^U_N - \mathbb{P}^L_n)\Delta_n = o_p(M^{-1/2}).$$

This proves the claimed first-order equivalence. In particular, $\widehat{\psi}_{\text{eff}}$ and $\widehat{\psi}_{\text{iso}}$ have the same asymptotic variance. □

Proof of corollary 2. For part (i), if $m_0 = \mu_0$ almost surely, then the asymptotic influence pair in theorem 2 is

$$D^L_{\mu_0}(X, Y) = \mu_0(X) - \psi_0 + \rho_0^{-1}\{Y - \mu_0(X)\}, \qquad D^U_{\mu_0}(\widetilde{X}) = \mu_0(\widetilde{X}) - \psi_0.$$

By proposition 5, this is the efficient influence pair in the unrestricted two-sample model. Therefore $\widehat{\psi}_{\text{iso}}$ is efficient for ψ_0 in the full two-sample experiment.

For part (ii), let

$$S := m(X), \qquad \eta_0(s) := \mathbb{E}_{P_0}[Y \mid S = s].$$

proposition 5 applied to the reduced two-sample experiment with scalar covariate S shows that the efficient influence pair in that reduced experiment is

$$D^L_{\eta_0}(S, Y) = \eta_0(S) - \psi_0 + \rho_0^{-1}\{Y - \eta_0(S)\}, \qquad D^U_{\eta_0}(\widetilde{S}) = \eta_0(\widetilde{S}) - \psi_0.$$

If η_0 is monotone increasing, then η_0 belongs to the isotonic class and

$$\mathbb{E}_{P_0}\left[\{Y - g(S)\}^2\right] = \mathbb{E}_{P_0}\left[\{Y - \eta_0(S)\}^2\right] + \mathbb{E}_{P_0}\left[\{\eta_0(S) - g(S)\}^2\right]$$

for every measurable g. Hence the population isotonic regression of Y on the score $S = m(X)$ is exactly $\eta_0(S)$, so

$$m_0(X) = \eta_0\{m(X)\} \qquad \text{almost surely.}$$

Therefore, the influence pair in theorem 2 coincides with the reduced-model efficient influence pair above, so $\widehat{\psi}_{\text{iso}}$ is efficient for ψ_0 in the reduced experiment indexed by $m(X)$. □

L Pooled i.i.d. Missing-Data Formulation

The joint two-sample setup can be embedded into a pooled i.i.d. model by introducing a sample-membership indicator $R \in \{0, 1\}$, where $R = 1$ denotes labeled observations and $R = 0$ denotes unlabeled observations. In that formulation one observes i.i.d. data $O = (X, R, RY)$ with $\mathbb{P}(R = 1) = \rho_0$, and the calibrated estimator becomes

$$\widehat{\psi}_{\text{cal}} = \mathbb{P}_M(m_n^\star) + \mathbb{P}_M\left[\frac{R}{\rho_0}\{Y - m_n^\star\}\right]$$

up to the same calibration identity as in the main text. The corresponding observed-data efficient influence function is

$$m_0(X) - \psi_0 + \frac{R}{\rho_0}\{Y - m_0(X)\},$$

which is the single-sample analogue of the efficient influence pair obtained by applying proposition 5 to the reduced model with scalar covariate $m(X)$. Thus the usual i.i.d. missing-data formulation is a convenient special case, but the main text keeps the two-dataset structure explicit.

M Additional empirical details

This appendix records the benchmark construction used in section 5.2. The main-text benchmark figure isolates three headline contrasts: PPI versus AIPW as raw-score baselines, clipped PPI++ versus AIPW-EM as efficiency-maximization variants, and `LinearCal`/`AutoCal` as the main calibration-based competitors.

We use the official `ppi_py` real-data mean-estimation examples and preserve their labeled-sample-size grids. The `forest` experiment uses $n \in \{50, 100, \ldots, 500\}$. The `galaxies` and semisupervised `census_income` benchmarks use $n \in \{50, 155, 261, 366, 472, 577, 683, 788, 894, 1000\}$. For each n, the remaining observations form the unlabeled sample. In the current draft we rerandomize this split 500 times at each n, always evaluating all estimators on the same split and taking the full-sample mean as the benchmark truth.

The estimator set mirrors the original PPI comparison wherever possible. In the paper-facing summaries, we report the labeled-only estimator, the imputation benchmark on binary-outcome tasks, PPI computed using the official `ppi_py` implementation, the classical AIPW estimator based on the

same prediction score, clipped PPI++, and the unclipped efficiency-maximized benchmark AIPW-EM. We then add five calibration-oriented comparators: linear calibration; smooth monotone spline calibration; Platt scaling on the binary-outcome benchmarks; isotonic calibration with a fixed minimum bin size of 10; and an adaptive `AutoCal` selector that chooses among AIPW, linear calibration, monotone spline calibration, and isotonic calibration by cross-validated empirical efficiency.

For the binary-outcome benchmarks, we also report Venn–Abers shrinkage. Platt scaling and Venn–Abers are omitted on `census_income` because that outcome is not binary. Including both PPI++ and AIPW-EM lets us separate the underlying efficiency-maximization idea from the specific `ppi_py` clipping rule for the tuning coefficient, while figure 5 isolates the fixed-calibration comparison among PPI, AIPW, `LinearCal`, `MonoSpline`, and `IsoCal`.

For the fixed calibration-based plug-in estimators, we fit the calibrator on the labeled pairs $\{(m(X_i), Y_i)\}_{i=1}^{n}$, transform both the labeled and unlabeled scores, and then compute the pooled plug-in mean together with the same semisupervised influence-style Wald interval used throughout the benchmark code. AIPW uses the same score but averages it over the pooled covariate sample before adding the labeled residual correction, whereas PPI uses only the unlabeled score average. In theory this retains less labeled-sample score information, though in the finite-sample benchmark results the empirical variance gap is often small and not uniform across all datasets. PPI++ and AIPW-EM both apply one-dimensional empirical efficiency maximization to the same score, but PPI++ inherits the official `ppi_py` clipping rule $\lambda \in [0, 1]$ whereas AIPW-EM does not.

Linear calibration regresses Y on an intercept and the score, then clips the fitted values to the range of predicted outcomes observed in the labeled data to prevent extrapolation on the unlabeled data. `MonoSpline` instead fits a smooth nondecreasing spline of Y on the score, using monotonicity constraints together with a mild roughness penalty, and then applies the same clipped pooled plug-in construction. Platt scaling fits a logistic regression of the binary outcome on a stabilized logit transform of the score, while the fixed-bin isotonic estimator fits a stepwise monotone calibration map and also clips to the labeled outcome range.

The `AutoCal` estimator is a selector rather than a new calibrator: in the experiments it compares AIPW, linear calibration, monotone spline calibration, and fixed-bin isotonic calibration using 20-fold cross-validation and the estimated influence-function variance as the selection criterion. To keep this step inexpensive when the unlabeled sample is very large, the foldwise criterion is evaluated on an unlabeled subsample of size $\min(N, 10n)$. The reported summaries are Monte Carlo averages of bias, empirical variance, MSE, interval coverage, and relative efficiency versus PPI across the repeated random splits. The complete numerical outputs, including dataset-level tables and per-metric plots, are generated automatically by the accompanying reproduction pipeline.

We also retain a supplementary screening sweep for the earlier score-decile gap construction. The screening compares the three public PPE-correctness evaluators (ArmoRM, Athene, Skywork) with the three reproduced tabular PPI benchmarks, always using the same top-versus-bottom-decile construction and the same small-sample calibration workflow. Table 2 helps explain why that decile-gap construction is less attractive for the introduction: the classic tabular benchmarks show much smaller incremental gains, ArmoRM is comparatively mild, and Skywork is too pathological.

Candidate	μ_A	μ_B	score_A	score_B	MSE drop	Note
Athene (PPE correctness)	0.790	0.212	1.751	-1.777	94%	Chosen intro anchor
ArmoRM (PPE correctness)	0.786	0.212	0.052	-0.053	62%	Benign real failure, but too mild for the main toy
Galaxies	0.973	0.011	0.981	0.006	-4%	Real benchmark, but smaller incremental gain
Census income	147977.895	395.886	106255.384	-110.489	8%	Real benchmark, but smaller incremental gain
Forest cover	0.630	0.018	0.727	0.013	4%	Real benchmark, but smaller incremental gain
Skywork (PPE correctness)	0.775	0.229	22.901	-23.555	100%	Very large gain, but driven by pathological raw scaling

Table 2: Supplementary screening sweep for the earlier score-decile gap construction. Each row defines arms A and B as the top and bottom raw-score deciles for that real candidate, reports the resulting human means and raw-score means, and summarizes the held-out squared-error drop from one-dimensional recalibration. The bolded row marks the strongest candidate under that decile-gap screening rule.

M.1 LLM evaluator breakdown

The PPE macro-averages in the main text hide substantial evaluator heterogeneity. In particular, the largest raw-proxy failures are driven primarily by Skywork and Athene, while ArmoRM is compar-

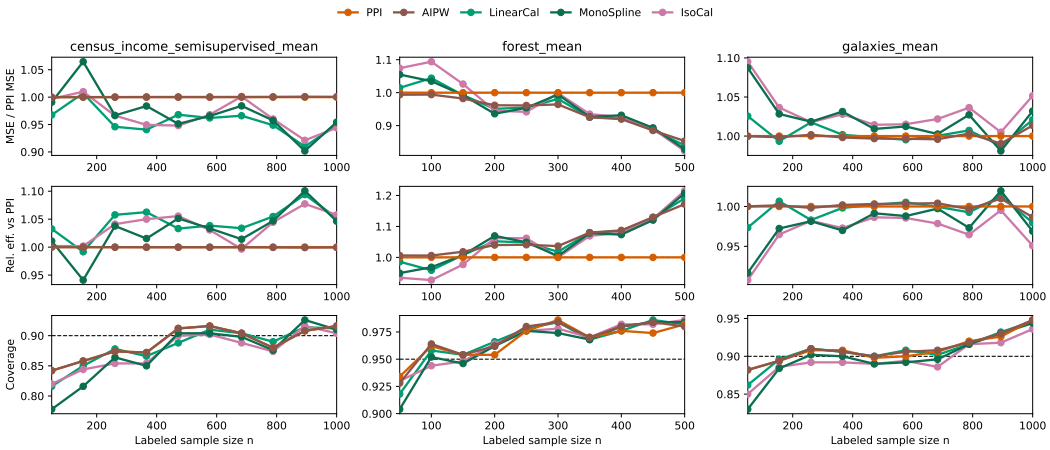

Figure 5: Calibration-focused appendix comparison across the reproduced PPI benchmarks, restricted to PPI, AIPW, `LinearCal`, `MonoSpline`, and `IsoCal`. The panels report normalized MSE relative to PPI, relative efficiency versus PPI, and coverage across labeled-sample-size regimes. This figure isolates the fixed-calibration comparison omitted from the main-text benchmark figure for readability.

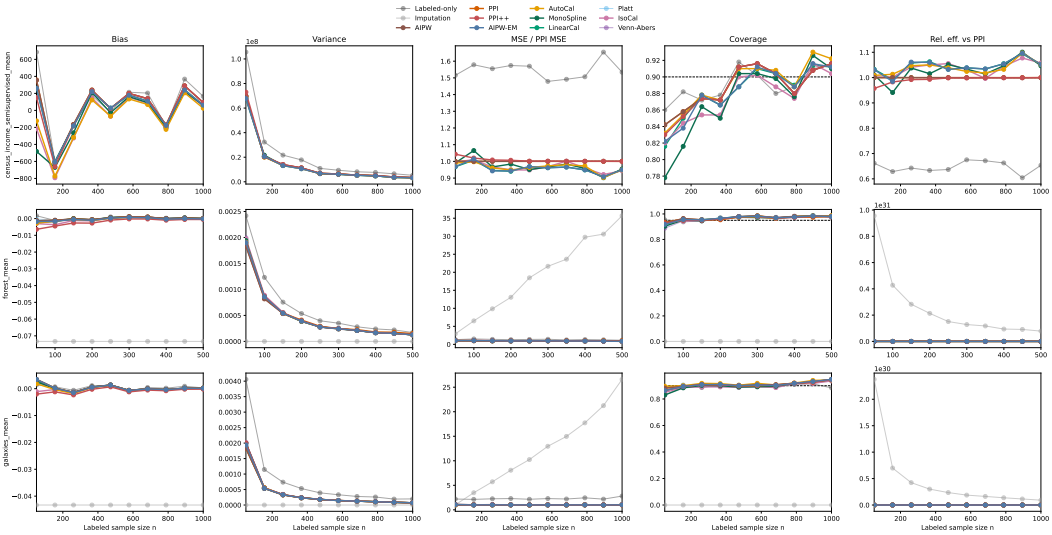

Figure 6: Full diagnostic grid for the reproduced PPI benchmarks, showing bias, empirical variance, normalized MSE relative to PPI, coverage, and relative efficiency versus PPI across labeled-sample-size regimes. In the normalized-MSE and relative-efficiency panels, the dashed horizontal line marks parity with PPI; in the coverage panels it marks the nominal target coverage $1 - \alpha$. `AutoCal`, `MonoSpline`, and `IsoCal` appear alongside PPI, AIPW, PPI++, AIPW-EM, and the fixed calibration rules, while Platt scaling and Venn–Abers appear only on the binary-outcome datasets where they are well defined.

atively benign and leaves all main-text estimators nearly tied. The calibration-based and efficiency-maximized methods are therefore best understood as safeguards against badly scaled reward-model margins, rather than gains that require every evaluator to be severely misaligned.

M.2 LLM ranking supplement

The PPE human-preference track also induces a downstream ranking problem across the selected target models. Using the same repeated labeled/unlabeled splits as in section 5.3, we therefore

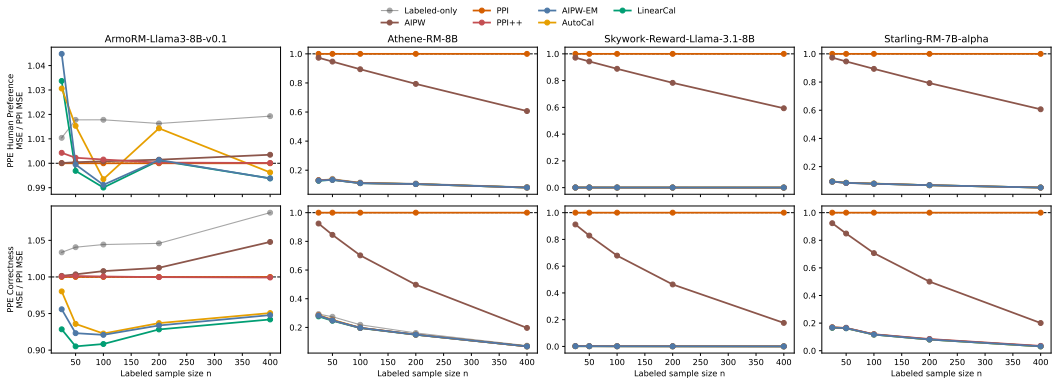

Figure 7: Appendix evaluator-specific PPE summary. Each panel repeats the PPE Human and PPE Correctness comparisons for one public evaluator. The large raw-proxy failures are concentrated in Skywork and Athene, while ArmoRM is much more benign.

summarize how well each estimator recovers the target-model ordering induced by the full-data human win shares. For each evaluator, labeled-sample size, replicate, and estimator, we rank the eight selected target models by estimated win share, compare that ordering against the ranking induced by the full-data targets, and then macro-average across evaluators. We report Spearman rank correlation, top-1 identification rate, and top-1 regret, defined as the difference in true win share between the best target model and the model selected by the estimator. The main pattern mirrors the mean-estimation results: calibrated and efficiency-maximized estimators substantially outperform raw PPI/AIPW, while remaining close to labeled-only at small n. At $n = 200$, `AutoCal` attains the highest average Spearman correlation, about 0.905; at $n = 400$, `LinearCal` gives the best overall ranking quality, with Spearman about 0.941 and top-1 identification about 0.646.

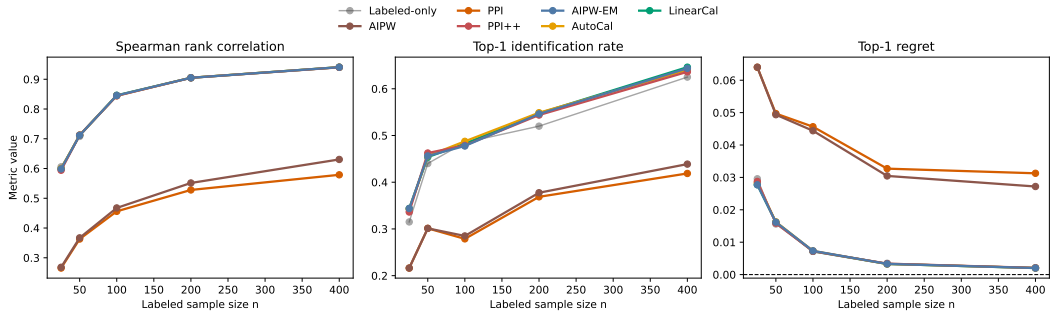

Figure 8: Appendix PPE-ranking supplement from the LLM benchmark, macro-averaged across the public evaluator models. The three panels report Spearman rank correlation, top-1 identification rate, and top-1 regret for recovering the ranking of the eight selected PPE target models from a small labeled sample plus evaluator scores.

N `Python` Code

Code availability. To preserve double-blind review, repository and documentation links are omitted from the submission PDF. This appendix instead gives a minimal anonymous implementation sketch that can be bundled with any supplementary artifact.

Below we also record the three main algorithmic summaries from the full paper, using the notation of the main text.

N.1 Algorithmic summaries

Below, we provide a minimal, self-contained implementation of the method introduced in this paper.

Algorithm 1 Isotonic-calibrated PPI

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $\hat{f} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \{Y_i - f(T_i)\}^2$
3: Set $m^\star_{n,\text{iso}}(X_i) = \hat{f}(T_i)$ and $m^\star_{n,\text{iso}}(\widetilde{X}_j) = \hat{f}(\widetilde{T}_j)$
4: Form $\widehat{\psi}_{\text{iso}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\text{iso}}(X_i) + \sum_{j=1}^N m^\star_{n,\text{iso}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = m^\star_{n,\text{iso}}(X_i) - \widehat{\psi}_{\text{iso}} + \rho_n^{-1}\{Y_i - m^\star_{n,\text{iso}}(X_i)\}$ and $\widehat{D}_j^U = m^\star_{n,\text{iso}}(\widetilde{X}_j) - \widehat{\psi}_{\text{iso}}$
6: Set $\widehat{\text{SE}}^2_{\text{iso}} = (n+N)^{-2}\{\sum_{i=1}^n (\widehat{D}_i^L)^2 + \sum_{j=1}^N (\widehat{D}_j^U)^2\}$
7: Return $\widehat{\psi}_{\text{iso}} \pm z_{1-\alpha/2}\widehat{\text{SE}}_{\text{iso}}$

Algorithm 2 Linearly calibrated PPI

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $(\hat{a}, \hat{b}) \in \arg\min_{a,b} \sum_{i=1}^n \{Y_i - (aT_i + b)\}^2$
3: Set $m^\star_{n,\text{lin}}(X_i) = \hat{a}T_i + \hat{b}$ and $m^\star_{n,\text{lin}}(\widetilde{X}_j) = \hat{a}\widetilde{T}_j + \hat{b}$
4: Form $\widehat{\psi}_{\text{lin}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\text{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\text{lin}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = m^\star_{n,\text{lin}}(X_i) - \widehat{\psi}_{\text{lin}} + \rho_n^{-1}\{Y_i - m^\star_{n,\text{lin}}(X_i)\}$ and $\widehat{D}_j^U = m^\star_{n,\text{lin}}(\widetilde{X}_j) - \widehat{\psi}_{\text{lin}}$
6: Set $\widehat{\text{SE}}^2_{\text{lin}} = (n+N)^{-2}\{\sum_{i=1}^n (\widehat{D}_i^L)^2 + \sum_{j=1}^N (\widehat{D}_j^U)^2\}$
7: Return $\widehat{\psi}_{\text{lin}} \pm z_{1-\alpha/2}\widehat{\text{SE}}_{\text{lin}}$

N.2 Isotonic regression via XGBoost

```
import numpy as np
import xgboost as xgb

def isoreg_with_xgboost(x, y, max_depth=15, min_child_weight=20, weights=None):
    """
    Fit a monotone calibrator f so that f(x) is nondecreasing in x.

    Parameters
    ----------
    x : array-like, shape (n,) or (n, 1)
        Prediction score used for calibration. In the PPI application this is
        usually the score m(X), so x is typically one-dimensional.
    y : array-like, shape (n,)
        Labeled outcomes.
    max_depth : int, default=15
        Maximum depth for the one-round XGBoost fit.
    min_child_weight : float, default=20
        Minimum total weight in a terminal node.
    weights : array-like, optional
        Optional observation weights.

    Returns
    -------
    predict_fn : callable
        Function mapping new x values to calibrated scores f(x).
    """
    x = np.asarray(x)
    y = np.asarray(y).reshape(-1)
    x = x.reshape(len(y), -1)

    data = xgb.DMatrix(data=x, label=y, weight=weights)
    params = {
        "max_depth": max_depth,
```

Algorithm 3 Linearly calibrated PPI with covariate adjustment

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$

2: Fit $(\hat{a}, \hat{\beta}, \hat{b}) \in \arg\min_{a,\beta,b} \sum_{i=1}^n \{Y_i - (a + X_i^\top \beta + bT_i)\}^2$

3: Set $m^\star_{n,\mathrm{lin}}(X_i) = \hat{a} + X_i^\top \hat{\beta} + \hat{b}T_i$ and $m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) = \hat{a} + \widetilde{X}_j^\top \hat{\beta} + \hat{b}\widetilde{T}_j$

4: Form $\widehat{\psi}_{\mathrm{lin}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\mathrm{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\mathrm{lin}}(\widetilde{X}_j)\}$

5: Set $\widehat{D}_i^L = m^\star_{n,\mathrm{lin}}(X_i) - \widehat{\psi}_{\mathrm{lin}} + \rho_n^{-1}\{Y_i - m^\star_{n,\mathrm{lin}}(X_i)\}$ and $\widehat{D}_j^U = m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) - \widehat{\psi}_{\mathrm{lin}}$

6: Set $\widehat{\mathrm{SE}}_{\mathrm{lin}}^2 = (n+N)^{-2}\{\sum_{i=1}^n (\widehat{D}_i^L)^2 + \sum_{j=1}^N (\widehat{D}_j^U)^2\}$

7: Return $\widehat{\psi}_{\mathrm{lin}} \pm z_{1-\alpha/2}\widehat{\mathrm{SE}}_{\mathrm{lin}}$

```
        "min_child_weight": min_child_weight,
        "monotone_constraints": "(" + ",".join(["1"] * x.shape[1]) + ")",
        "eta": 1.0,
        "gamma": 0.0,
        "lambda": 0.0,
        "objective": "reg:squarederror",
        "verbosity": 0,
    }
    iso_fit = xgb.train(params=params, dtrain=data, num_boost_round=1)

    def predict_fn(x_new):
        x_new = np.asarray(x_new)
        if x_new.ndim == 1:
            x_new = x_new.reshape(-1, 1)
        data_pred = xgb.DMatrix(data=x_new)
        return iso_fit.predict(data_pred)

    return predict_fn
```

N.3 Isotonic-calibrated plug-in

```
import numpy as np
from statistics import NormalDist


def _resolve_scores(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
):
    Y_labeled = np.asarray(Y_labeled).reshape(-1)

    if score_fn is not None:
        if X_labeled is None or X_unlabeled is None:
            raise ValueError(
                "If score_fn is provided, then X_labeled and X_unlabeled "
                "must also be provided."
            )
        score_labeled = score_fn(X_labeled)
        score_unlabeled = score_fn(X_unlabeled)
    elif score_labeled is None or score_unlabeled is None:
        raise ValueError(
            "Provide either score_fn together with X_labeled/X_unlabeled, "
            "or provide score_labeled and score_unlabeled."
        )

    score_labeled = np.asarray(score_labeled).reshape(-1)
```

```
    score_unlabeled = np.asarray(score_unlabeled).reshape(-1)

    if len(score_labeled) != len(Y_labeled):
        raise ValueError("Y_labeled and score_labeled must have the same length.")

    return Y_labeled, score_labeled, score_unlabeled


def isotonic_calibrated_plugin(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
    alpha=0.05,
    min_obs=20,
):
    Y_labeled, score_labeled, score_unlabeled = _resolve_scores(
        Y_labeled=Y_labeled,
        X_labeled=X_labeled,
        X_unlabeled=X_unlabeled,
        score_fn=score_fn,
        score_labeled=score_labeled,
        score_unlabeled=score_unlabeled,
    )

    n = len(Y_labeled)
    N = len(score_unlabeled)
    M = n + N
    rho = n / M

    calibrator = isoreg_with_xgboost(
        x=score_labeled,
        y=Y_labeled,
        min_child_weight=min_obs,
    )

    m_labeled = calibrator(score_labeled)
    m_unlabeled = calibrator(score_unlabeled)

    psi_hat = (n * m_labeled.mean() + N * m_unlabeled.mean()) / M

    if_labeled = m_labeled - psi_hat + (Y_labeled - m_labeled) / rho
    if_unlabeled = m_unlabeled - psi_hat
    var_hat = (
        rho * np.var(if_labeled, ddof=1)
        + (1 - rho) * np.var(if_unlabeled, ddof=1)
    )
    se_hat = np.sqrt(var_hat / M)

    z = NormalDist().inv_cdf(1 - alpha / 2)
    ci = (psi_hat - z * se_hat, psi_hat + z * se_hat)

    return {
        "estimate": psi_hat,
        "standard_error": se_hat,
        "confidence_interval": ci,
    }


# Example:
# fit = isotonic_calibrated_plugin(
#     Y_labeled=Y,
#     X_labeled=X,
#     X_unlabeled=X_tilde,
```

```
#     score_fn=m,
# )
```

N.4 Linear calibration

```
import numpy as np
from statistics import NormalDist


def _resolve_scores(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
):
    Y_labeled = np.asarray(Y_labeled).reshape(-1)

    if score_fn is not None:
        if X_labeled is None or X_unlabeled is None:
            raise ValueError(
                "If score_fn is provided, then X_labeled and X_unlabeled "
                "must also be provided."
            )
        score_labeled = score_fn(X_labeled)
        score_unlabeled = score_fn(X_unlabeled)
    elif score_labeled is None or score_unlabeled is None:
        raise ValueError(
            "Provide either score_fn together with X_labeled/X_unlabeled, "
            "or provide score_labeled and score_unlabeled."
        )

    score_labeled = np.asarray(score_labeled).reshape(-1)
    score_unlabeled = np.asarray(score_unlabeled).reshape(-1)

    if len(score_labeled) != len(Y_labeled):
        raise ValueError("Y_labeled and score_labeled must have the same length.")

    return Y_labeled, score_labeled, score_unlabeled


def linear_calibrated_plugin(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
    alpha=0.05,
):
    Y_labeled, score_labeled, score_unlabeled = _resolve_scores(
        Y_labeled=Y_labeled,
        X_labeled=X_labeled,
        X_unlabeled=X_unlabeled,
        score_fn=score_fn,
        score_labeled=score_labeled,
        score_unlabeled=score_unlabeled,
    )

    n = len(Y_labeled)
    N = len(score_unlabeled)
    M = n + N
    rho = n / M

    X_design = np.column_stack([np.ones(n), score_labeled])
```

```
    beta_hat, _, _, _ = np.linalg.lstsq(X_design, Y_labeled, rcond=None)

    def calibrator(score):
        score = np.asarray(score).reshape(-1)
        return beta_hat[0] + beta_hat[1] * score

    m_labeled = calibrator(score_labeled)
    m_unlabeled = calibrator(score_unlabeled)

    psi_hat = (n * m_labeled.mean() + N * m_unlabeled.mean()) / M

    if_labeled = m_labeled - psi_hat + (Y_labeled - m_labeled) / rho
    if_unlabeled = m_unlabeled - psi_hat
    var_hat = (
        rho * np.var(if_labeled, ddof=1)
        + (1 - rho) * np.var(if_unlabeled, ddof=1)
    )
    se_hat = np.sqrt(var_hat / M)

    z = NormalDist().inv_cdf(1 - alpha / 2)
    ci = (psi_hat - z * se_hat, psi_hat + z * se_hat)

    return {
        "estimate": psi_hat,
        "standard_error": se_hat,
        "confidence_interval": ci,
    }


# Example:
# fit = linear_calibrated_plugin(
#     Y_labeled=Y,
#     score_labeled=mX,
#     score_unlabeled=mX_tilde,
# )
```

Dataset	n	N	Estimator	Bias	Variance	MSE	Coverage	RelEff
census_income_semisupervised_mean	50	380041	AIPW	356.4905	69836665.4410	69963750.9366	0.842	1.000
census_income_semisupervised_mean	50	380041	AIPW-EM	260.9833	67728426.6042	67796538.8913	0.822	1.031
census_income_semisupervised_mean	50	380041	AutoCal	-123.2501	69242944.1124	69258134.7078	0.832	1.009
census_income_semisupervised_mean	50	380041	Labeled-only	685.3532	105558402.2427	106028111.3137	0.860	0.662
census_income_semisupervised_mean	50	380041	IsoCal	-215.3832	69697400.2434	69743790.1536	0.820	1.002
census_income_semisupervised_mean	50	380041	LinearCal	260.9294	67622601.4006	67690685.5430	0.816	1.033
census_income_semisupervised_mean	50	380041	MonoSpline	-482.0477	69082916.1270	69315286.1508	0.778	1.011
census_income_semisupervised_mean	50	380041	PPI	356.4473	69834865.9486	69961920.5977	0.842	1.000
census_income_semisupervised_mean	50	380041	PPI++	150.8664	72953053.8328	72975814.4980	0.830	0.957
census_income_semisupervised_mean	577	379514	AIPW	199.8136	6353884.5833	6393810.0648	0.916	1.000
census_income_semisupervised_mean	577	379514	AIPW-EM	177.6067	6114918.2974	6146462.4536	0.912	1.039
census_income_semisupervised_mean	577	379514	AutoCal	134.3746	6201029.6459	6219086.1828	0.910	1.024
census_income_semisupervised_mean	577	379514	Labeled-only	213.2465	9407595.6062	9453069.6946	0.902	0.675
census_income_semisupervised_mean	577	379514	IsoCal	155.0503	6162964.7883	6187005.3977	0.902	1.031
census_income_semisupervised_mean	577	379514	LinearCal	179.8005	6116832.9552	6149161.1594	0.910	1.038
census_income_semisupervised_mean	577	379514	MonoSpline	162.1952	6146069.6864	6172376.9541	0.904	1.033
census_income_semisupervised_mean	577	379514	PPI	199.7932	6351836.6166	6391753.9369	0.916	1.000
census_income_semisupervised_mean	577	379514	PPI++	198.7409	6362047.6345	6401545.5819	0.916	0.998
census_income_semisupervised_mean	1000	379091	AIPW	92.1091	3462051.4638	3470535.5425	0.916	0.999
census_income_semisupervised_mean	1000	379091	AIPW-EM	61.4705	3288235.4312	3292014.0588	0.912	1.052
census_income_semisupervised_mean	1000	379091	AutoCal	25.8207	3289467.9320	3290134.6381	0.922	1.052
census_income_semisupervised_mean	1000	379091	Labeled-only	165.4763	5293296.2822	5320678.6872	0.914	0.654
census_income_semisupervised_mean	1000	379091	IsoCal	51.2697	3271622.4413	3274251.0223	0.904	1.058
census_income_semisupervised_mean	1000	379091	LinearCal	60.7501	3287796.5296	3291487.1058	0.912	1.052
census_income_semisupervised_mean	1000	379091	MonoSpline	60.9626	3306575.6859	3310292.1289	0.910	1.046
census_income_semisupervised_mean	1000	379091	PPI	91.9155	3460071.7259	3468520.1894	0.916	1.000
census_income_semisupervised_mean	1000	379091	PPI++	91.9155	3460071.7259	3468520.1894	0.916	1.000
forest_mean	50	1546	AIPW	-0.0012	0.0018	0.0018	0.928	1.006
forest_mean	50	1546	AIPW-EM	-0.0018	0.0019	0.0019	0.918	0.969
forest_mean	50	1546	AutoCal	-0.0028	0.0019	0.0019	0.924	0.965
forest_mean	50	1546	Labeled-only	0.0017	0.0024	0.0024	0.954	0.762
forest_mean	50	1546	Imputation	-0.0733	0.0000	0.0054	0.000	9564441057429143054280466366464.000
forest_mean	50	1546	IsoCal	-0.0030	0.0020	0.0020	0.930	0.934
forest_mean	50	1546	LinearCal	-0.0016	0.0019	0.0019	0.918	0.986
forest_mean	50	1546	MonoSpline	-0.0021	0.0019	0.0019	0.904	0.949
forest_mean	50	1546	Platt	-0.0003	0.0019	0.0019	0.900	0.948
forest_mean	50	1546	PPI	-0.0013	0.0018	0.0018	0.934	1.000
forest_mean	50	1546	PPI++	-0.0064	0.0019	0.0019	0.928	0.969
forest_mean	50	1546	Venn-Abers	-0.0027	0.0020	0.0020	0.888	0.923
forest_mean	300	1296	AIPW	0.0011	0.0002	0.0002	0.984	1.036
forest_mean	300	1296	AIPW-EM	0.0007	0.0002	0.0002	0.982	1.012
forest_mean	300	1296	AutoCal	0.0006	0.0002	0.0002	0.978	1.002
forest_mean	300	1296	Labeled-only	0.0007	0.0003	0.0004	0.968	0.706
forest_mean	300	1296	Imputation	-0.0733	0.0000	0.0054	0.000	1281441200512508097500322725888.000
forest_mean	300	1296	IsoCal	0.0010	0.0002	0.0002	0.978	1.000
forest_mean	300	1296	LinearCal	0.0010	0.0002	0.0002	0.984	1.018
forest_mean	300	1296	MonoSpline	0.0010	0.0002	0.0002	0.974	1.004
forest_mean	300	1296	Platt	0.0010	0.0002	0.0002	0.982	1.020
forest_mean	300	1296	PPI	0.0011	0.0002	0.0002	0.986	1.000
forest_mean	300	1296	PPI++	-0.0002	0.0002	0.0002	0.980	0.991
forest_mean	300	1296	Venn-Abers	0.0011	0.0002	0.0002	0.978	1.015
forest_mean	500	1096	AIPW	0.0001	0.0001	0.0001	0.980	1.172
forest_mean	500	1096	AIPW-EM	-0.0002	0.0001	0.0001	0.980	1.188
forest_mean	500	1096	AutoCal	-0.0003	0.0001	0.0001	0.986	1.180
forest_mean	500	1096	Labeled-only	0.0001	0.0002	0.0002	0.972	0.879
forest_mean	500	1096	Imputation	-0.0733	0.0000	0.0054	0.000	7819091278107866750853379522 56.000
forest_mean	500	1096	IsoCal	-0.0000	0.0001	0.0001	0.986	1.215
forest_mean	500	1096	LinearCal	0.0001	0.0001	0.0001	0.982	1.191
forest_mean	500	1096	MonoSpline	0.0000	0.0001	0.0001	0.984	1.207
forest_mean	500	1096	Platt	0.0001	0.0001	0.0001	0.984	1.203
forest_mean	500	1096	PPI	0.0000	0.0002	0.0002	0.982	1.000
forest_mean	500	1096	PPI++	-0.0006	0.0001	0.0001	0.978	1.179
forest_mean	500	1096	Venn-Abers	-0.0001	0.0001	0.0001	0.978	1.206
galaxies_mean	50	16693	AIPW	0.0035	0.0018	0.0018	0.882	1.000
galaxies_mean	50	16693	AIPW-EM	0.0035	0.0019	0.0020	0.864	0.944
galaxies_mean	50	16693	AutoCal	0.0019	0.0018	0.0018	0.896	0.997
galaxies_mean	50	16693	Labeled-only	0.0014	0.0041	0.0041	0.890	0.451
galaxies_mean	50	16693	Imputation	-0.0433	0.0000	0.0019	0.000	2376289106442477113652488962048.000
galaxies_mean	50	16693	IsoCal	-0.0011	0.0020	0.0020	0.850	0.908
galaxies_mean	50	16693	LinearCal	0.0030	0.0019	0.0019	0.862	0.973
galaxies_mean	50	16693	MonoSpline	0.0026	0.0020	0.0020	0.830	0.917
galaxies_mean	50	16693	Platt	0.0033	0.0019	0.0019	0.856	0.966
galaxies_mean	50	16693	PPI	0.0035	0.0018	0.0018	0.882	1.000
galaxies_mean	50	16693	PPI++	-0.0020	0.0020	0.0020	0.864	0.914
galaxies_mean	50	16693	Venn-Abers	0.0022	0.0020	0.0020	0.830	0.907
galaxies_mean	577	16166	AIPW	-0.0006	0.0001	0.0001	0.906	1.004
galaxies_mean	577	16166	AIPW-EM	-0.0007	0.0001	0.0001	0.906	1.005
galaxies_mean	577	16166	AutoCal	-0.0008	0.0001	0.0001	0.918	0.989
galaxies_mean	577	16166	Labeled-only	-0.0011	0.0003	0.0003	0.910	0.436
galaxies_mean	577	16166	Imputation	-0.0433	0.0000	0.0019	0.000	1873962492481163727822158 56128.000
galaxies_mean	577	16166	IsoCal	-0.0007	0.0001	0.0001	0.894	0.986
galaxies_mean	577	16166	LinearCal	-0.0007	0.0001	0.0001	0.908	1.005
galaxies_mean	577	16166	MonoSpline	-0.0006	0.0001	0.0001	0.892	0.988
galaxies_mean	577	16166	Platt	-0.0007	0.0001	0.0001	0.896	1.002
galaxies_mean	577	16166	PPI	-0.0006	0.0001	0.0001	0.900	1.000
galaxies_mean	577	16166	PPI++	-0.0012	0.0001	0.0001	0.902	0.988
galaxies_mean	577	16166	Venn-Abers	-0.0006	0.0001	0.0001	0.892	0.986
galaxies_mean	1000	15743	AIPW	0.0002	0.0001	0.0001	0.948	0.987
galaxies_mean	1000	15743	AIPW-EM	0.0002	0.0001	0.0001	0.948	0.980
galaxies_mean	1000	15743	AutoCal	0.0001	0.0001	0.0001	0.944	0.977
galaxies_mean	1000	15743	Labeled-only	0.0003	0.0002	0.0002	0.884	0.357
galaxies_mean	1000	15743	Imputation	-0.0433	0.0000	0.0019	0.000	9146652808341511999670190 0800.000
galaxies_mean	1000	15743	IsoCal	0.0001	0.0001	0.0001	0.936	0.951
galaxies_mean	1000	15743	LinearCal	0.0002	0.0001	0.0001	0.946	0.980
galaxies_mean	1000	15743	MonoSpline	0.0002	0.0001	0.0001	0.944	0.969
galaxies_mean	1000	15743	Platt	0.0002	0.0001	0.0001	0.942	0.979
galaxies_mean	1000	15743	PPI	0.0002	0.0001	0.0001	0.948	1.000
galaxies_mean	1000	15743	PPI++	-0.0002	0.0001	0.0001	0.946	0.977
galaxies_mean	1000	15743	Venn-Abers	0.0002	0.0001	0.0001	0.942	0.952

Table 3: Representative numerical summary at the smallest, middle, and largest labeled sample sizes for each reproduced benchmark. The reported quantities are Monte Carlo bias, empirical variance of the point estimator, MSE, Wald-interval coverage, and relative efficiency versus PPI for the displayed benchmark set, including PPI++, AIPW-EM, `AutoCal`, `MonoSpline`, and `IsoCal`.

n	Estimator	Spearman	Top-1	Top-1 regret
25	Labeled-only	0.606	0.315	0.030
25	PPI	0.265	0.216	0.064
25	AIPW	0.268	0.216	0.064
25	PPI++	0.594	0.336	0.029
25	AIPW-EM	0.599	0.344	0.028
25	LinearCal	0.599	0.344	0.028
25	AutoCal	0.600	0.343	0.028
200	Labeled-only	0.905	0.520	0.004
200	PPI	0.528	0.369	0.033
200	AIPW	0.551	0.378	0.030
200	PPI++	0.905	0.544	0.003
200	AIPW-EM	0.905	0.546	0.003
200	LinearCal	0.905	0.548	0.003
200	AutoCal	0.905	0.549	0.003
400	Labeled-only	0.940	0.625	0.002
400	PPI	0.579	0.419	0.031
400	AIPW	0.631	0.439	0.027
400	PPI++	0.940	0.636	0.002
400	AIPW-EM	0.941	0.644	0.002
400	LinearCal	0.941	0.646	0.002
400	AutoCal	0.941	0.640	0.002

Table 4: Appendix PPE-ranking summary at selected labeled sample sizes from the LLM benchmark. Spearman and top-1 are larger-is-better, while top-1 regret is smaller-is-better.

NeurIPS Paper Checklist

1. **Claims. Yes.** The abstract and introduction state the paper's main claims, and the paper distinguishes between exact representation results, asymptotic theory, and empirical findings; see Sections 2–5.
2. **Limitations. Yes.** The conclusion and theory sections state the paper's main scope limitations: the strongest guarantees use the known-ρ_n two-sample design and a stable second-stage calibrator; see Section 6 and Appendix B.
3. **Theory, assumptions, and proofs. Yes.** The main assumptions are stated in Sections 2 and 4, with complete proofs collected in Appendices G, H, I, J, and K.
4. **Experimental results. Yes.** The main experimental takeaways appear in Section 5, with benchmark construction and additional diagnostics deferred to Appendix M.
5. **Open access to data and code. No.** This anonymous submission does not include a public repository or packaged artifact; Appendix N provides representative implementation sketches, and benchmark sources are cited in the text.
6. **Experimental setting and details. Yes.** The simulation design, benchmark protocol, estimator set, and evaluation summaries are described in Section 5 and Appendix M.
7. **Experimental statistical significance. Yes.** The experiments report Monte Carlo variability, interval coverage, and efficiency comparisons, and the inference procedures are tied to the Wald theory in Section 4; see also Appendix C.
8. **Experiments compute resources. No.** The paper does not currently report wall-clock time, hardware, or aggregate compute usage for the simulation and benchmark runs.
9. **Code of ethics. Yes.** The paper is a methodological contribution in semisupervised statistical inference, and we are not aware of any aspect of the work that conflicts with the NeurIPS Code of Ethics.
10. **Broader impacts. N/A.** The paper develops estimation and inference methodology rather than a deployed decision system, though misuse of semisupervised estimators in high-stakes applications remains a general concern.
11. **Safeguards. N/A.** The paper does not introduce a system that generates content or interacts with users, and it does not release new high-risk artifacts requiring special safeguards.
12. **Licenses for existing assets. No.** The current draft cites the benchmark sources and related software ecosystems but does not yet include a consolidated license/version table for all reused assets.
13. **New assets. N/A.** The paper does not introduce a new dataset, benchmark, or foundation model.
14. **Crowdsourcing and research with human subjects. N/A.** The paper does not involve human-subject experiments or crowdsourcing.